The Impact of Job Displacement on Health Insurance Status in the Post-ACA Era*

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Abstract

This paper examines how job displacement affects health insurance coverage after the Affordable Care Act (ACA), whether alternative coverage sources offset losses of employer-sponsored health insurance (ESHI), and how these effects compare with those before the ACA. I use the 2008, 2014, and 2018 panels of the Survey of Income and Program Participation and a two-stage difference-in-differences design. In the post-ACA period, displacement reduces own ESHI coverage by 11.7 percentage points and increases uninsurance by 7.0 points one month after job loss. Medicaid enrollment rises gradually, particularly among lower-income workers, while directly purchased and spousal private coverage show small average responses. In expansion states, a sustained Medicaid increase is observed in 2014–2015 but not in 2017–2018. Comparing displacement effects before and after the ACA shows smaller losses of own ESHI and smaller increases in uninsurance after the ACA, with no comparably clear differences in responses through alternative coverage sources. These findings point to a continuing challenge in maintaining health insurance coverage immediately after job loss in the post-ACA period.

Keywords: Health Insurance, Job Displacement, the Affordable Care Act

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1 Introduction

Most non-elderly Americans receive health insurance through their employers. Such insurance is called employer-sponsored health insurance (ESHI). But linking insurance coverage to employment creates an inherent risk. Workers who lose their jobs involuntarily, a shock commonly referred to as job displacement, face not only earnings losses but also potential loss of ESHI. In fact, prior to the Affordable Care Act (ACA), various research documented substantial declines in insurance coverage following job displacement (Lin (2005); Jolly and Phelan (2015); Schaller and Stevens (2015); Jolly and Phelan (2017); East and Simon (2024)).

The ACA introduced reforms that could alleviate these coverage losses. The ACA transformed health insurance options for displaced workers through three main mechanisms. First, it established health insurance exchanges/marketplaces (HIX) with income-based premium subsidies for qualified individuals. Second, it expanded Medicaid eligibility in participating states to cover more low-income adults. Third, it implemented employer mandates that require larger employers to offer affordable health insurance. These reforms created multiple pathways for maintaining coverage outside employment while reinforcing the employer-sponsored insurance system.

This paper examines the impact of job displacement on health insurance coverage in the post-ACA environment. I address three questions: (1) To what extent do displaced workers lose ESHI coverage after job displacement? (2) How effectively do alternative coverage sources, particularly those expanded under the ACA, compensate for lost ESHI? (3) How do the magnitude of insurance losses and responses across coverage sources compare before and after the ACA? Since job displacement is a common economic shock affecting millions of Americans annually, understanding whether transitions from employer-sponsored coverage to alternative sources occur smoothly is important.

To answer these questions, I estimate the causal impact of job displacement on various insurance coverage rates using a difference-in-differences design. The approach compares

changes in insurance coverage around displacement between workers who are displaced earlier and those not yet displaced. The analysis uses pooled monthly data from the 2014 and 2018 panels of the Survey of Income and Program Participation (SIPP), covering 2014–2015 and 2017–2018, respectively. SIPP provides detailed information on both labor market status and health insurance coverage. I implement the two-stage difference-in-differences estimator developed by Gardner et al. (2024). For the pre/post comparison, I use the 2008 SIPP panel covering 2011–2012 as the pre-ACA benchmark and compare coverage responses across periods at the wave level.

The post-ACA estimates show that job displacement still causes an immediate loss of ESHI and an increase in uninsurance. One month after job loss, own-name ESHI coverage falls by 11.7 percentage points and uninsurance rises by 7.0 points. Medicaid enrollment rises gradually, while own-name directly purchased private insurance (DPHI) and spouse's private coverage show small average responses, suggesting a limited role in offsetting ESHI losses. The Medicaid response varies by income, state expansion status, and period. It is larger among lower-income workers. In expansion states, the increase is sustained in the 2014 SIPP panel, whereas no comparable sustained response is observed in the 2018 panel. Although ESHI losses and increases in uninsurance diminish over subsequent months, these findings suggest that maintaining coverage immediately after displacement remains a challenge.

The pre/post comparison shows smaller losses of own ESHI and smaller increases in uninsurance after the ACA. The clearest difference across coverage sources is in own ESHI, whose decline is smaller both in absolute terms and relative to employment losses, while Medicaid and other private coverage responses show no comparably clear differences. These patterns suggest that the adverse coverage consequences of displacement have become less severe, although the comparison does not identify the causal effect of the ACA.

This study contributes to the literature on job displacement and health insurance.

Previous research has examined the coverage consequences of job loss in pre-ACA settings (Gruber and Madrian, 1997; Kosteas and Renna, 2009; Jolly and Phelan, 2015; Schaller and Stevens, 2015; Jolly and Phelan, 2017), with East and Simon (2024) showing that declines in employer-provided insurance were partially offset by public insurance take-up. This paper extends the analysis to the post-ACA environment, where marketplace plans and expanded Medicaid offer new coverage pathways, and traces coverage responses by source and policyholder over the twelve months following job loss. It also compares these responses before and after the ACA to assess how the coverage consequences of displacement differ across periods.[1]

The remainder of this paper proceeds as follows. Section 2 describes the institutional background and data. Section 3 presents the empirical strategy. Section 4 addresses the first two research questions by examining post-ACA displacement effects and responses across coverage sources. Section 5 addresses the third question by comparing displacement effects before and after the ACA. Section 6 discusses the findings. Section 7 concludes.

2 Background and data

2.1 Institutional background

ESHI has long been the primary source of health insurance coverage for non-elderly Americans. This link between employment and insurance coverage creates an inherent risk for displaced workers because they face a potential loss of ESHI in addition to income loss. In this section, I describe several options for displaced workers to maintain insurance coverage and how these options evolved following the implementation of ACA in 2014.

[1] In concurrent work developed independently of this paper, Schaller and Zerpa (2026) also document smaller insurance losses after the ACA. Section 6 discusses the relationship between the two studies and their differences.

Continuation Coverage of ESHI under COBRA The Consolidated Omnibus Budget Reconciliation Act of 1985 (COBRA) allows displaced workers to temporarily continue their ESHI coverage. COBRA applies to firms with 20 or more employees and provides up to 18 months of continuation coverage after job termination. Eligible individuals have 60 days to elect coverage [2]. Despite availability, this option sees limited use. In 2017, approximately 130,000 unemployed non-elderly adults maintained COBRA coverage out of 11.5 million unemployed non-elderly adults (Pollitz et al. (2020)). Cost may contribute to the limited use of COBRA. When displaced workers enroll in continuation ESHI under COBRA, they must pay both employee and employer premium portions plus administrative fees (Elisabeth Thames Askin (2024)). This cost can burden displaced workers who have lost income.

ESHI coverage under employer mandate The ACA established an employer mandate that requires firms with 50 or more full-time equivalent employees to provide affordable health insurance to their full-time employees or pay fines. The employer mandate could affect displaced workers through two channels. First, it could increase the likelihood that displaced workers find new employers offering ESHI. This may accelerate re-insurance through new employment. Second, by increasing ESHI offer rates among larger employers, it could expand the pool of displaced workers with COBRA eligibility, since only workers who had ESHI can access COBRA benefits.

Spousal coverage through HIPAA The Health Insurance Portability and Accountability Act (HIPAA) enables displaced workers to enroll in a spouse's employer plan outside regular enrollment periods. This requires enrollment within 30 days of losing ESHI. HIPAA allows family members to make independent coverage decisions, potentially splitting coverage across different plans based on their own needs [3].

[2] See Pollitz et al. (2020) for details on COBRA eligibility and enrollment rules.
[3] See U.S. Department of Labor (2025) for details on special enrollment rights under HIPAA.

Directly purchased private insurance When displaced workers lose ESHI, they also have the option to purchase private health insurance directly from insurance carriers or brokers. Prior to the ACA, this option was rarely utilized, with only about 6% of non-elderly Americans obtaining coverage through this channel between 2008-2013[4]. Insurers sometimes denied coverage based on pre-existing conditions or imposed differential premiums. Additionally, individual market premiums typically exceeded ESHI premiums due to various reasons such as absence of employer premium contributions and limited risk pooling [5].

The ACA transformed the individual insurance market through three key mechanisms: (i) the ACA established health insurance exchanges (marketplaces) where individuals can compare and purchase standardized plans. Loss of job-based coverage qualifies as a "qualifying life event" that triggers a Special Enrollment Period, allowing workers to buy a marketplace plan within 60 days of ESHI loss. (ii) the ACA provides income-based premium subsidies through Premium Tax Credits. During the study periods of 2014–2015 and 2017–2018, these subsidies were available to marketplace enrollees with household incomes between 100 and 400% of FPL, with no subsidies above that threshold. In addition, Cost-Sharing Reductions provide additional assistance with out-of-pocket expenses for lower-income enrollees (100-250% FPL) in specific plans. (iii) the ACA prohibits coverage denials based on pre-existing conditions. These reforms were designed to make directly purchased private insurance more accessible and affordable for individuals without ESHI. The ACA may also shift the relative attractiveness of COBRA, as subsidized marketplace plans are typically more affordable than unsubsidized COBRA coverage[6][7].

[4]KFF estimates based on the 2008-2023 American Community Survey, 1-Year Estimates.

[5]See KFF (2024) for details on the advantages of ESHI over individual market coverage.

[6]Some workers may still prefer COBRA continuation coverage because they value the provider network of the original ESHI.

[7]Alternative individual market options include off-marketplace plans ineligible for subsidies such as Short-Term Limited Duration Insurance, which typically offers less comprehensive but more affordable coverage.

Medicaid Expansion The ACA's Medicaid expansion extended public coverage eligibility for low-income adults. Pre-ACA, Medicaid eligibility usually excluded childless non-disabled adults regardless of their income. The ACA aimed to expand eligibility to nearly all non-elderly adults with incomes up to 138% FPL. However, the 2012 Supreme Court decision made expansion optional for states. As of early 2025, 41 states including DC have implemented expansion, while 10 have not. In expansion states, workers with post-displacement income below 138% FPL typically qualify for Medicaid. In non-expansion states, displaced workers can fall into a "coverage gap" if their incomes exceed their state's eligibility threshold but remain below 100% FPL. In this case, they are ineligible for both Medicaid and ACA marketplace subsidies. Workers with incomes above 100% FPL in non-expansion states can access marketplace subsidies but face higher costs than they would under Medicaid [8].

2.2 Data and sample selection

For the analysis, this paper uses pooled data from the 2014 and 2018 panels of the Survey of Income and Program Participation (SIPP). The 2014 panel consists of four waves covering 2013 (wave 1) through 2016 (wave 4), and the 2018 panel consists of four waves covering 2017 (wave 1) through 2020 (wave 4).[9] SIPP fits well with the purpose of this paper for two reasons. It is nationally representative, and each wave contains monthly information on health insurance coverage, employment status, labor market outcomes, and demographics. This monthly structure provides precise measurement of insurance transitions after job displacement.

I restrict the sample to working-age adults (26-59 years old). The sample excludes those reporting retirement during the study period, the self-employed, and Medicare

[8] See KFF (2025) for details on state-by-state Medicaid expansion status.

[9] The separate 2019 SIPP panel was not followed in subsequent years and therefore does not provide the longitudinal observations required for this design. The Census Bureau also advises against analyzing the 2019 Panel Wave 1 cases alone because data collection problems led it to prioritize continued follow-up of the 2018 panel (U.S. Census Bureau, *2020 SIPP Users' Guide*, p. 7).

and military health insurance beneficiaries[10]. The sample includes only respondents observed continuously from 2014 (wave 2) through 2015 (wave 3) in the 2014 panel, or from 2017 (wave 1) through 2018 (wave 2) in the 2018 panel. I limit each estimation window to two years to retain more observations while requiring a balanced panel. I focus specifically on workers who experienced displacement at some point between 2014 (wave 2) and 2016 (wave 4) in the 2014 panel, or between 2017 (wave 1) and 2019 (wave 3) in the 2018 panel, as never-displaced workers may differ systematically from displaced workers (Krolikowski (2018))[11]. Workers displaced during 2013 (wave 1) are excluded from the 2014-panel sample to avoid "always-treated" cases in the estimation framework. Workers displaced during 2020 (wave 4) are excluded from the 2018-panel sample because displacement during the COVID-19 pandemic may differ substantially from displacement in the preceding years. Note that never-displaced workers, who are excluded from the sample, are distinct from never-treated workers. Never-treated workers experienced displacement during 2016 (wave 4) in the 2014 panel or during 2019 (wave 3) in the 2018 panel, outside their respective estimation windows, and constitute part of the control group alongside not-yet-treated observations from later cohorts. Appendix Figure A1 summarizes the sample construction and estimation window. Detailed discussion on the construction of treatment and control groups will be provided in Section 3.

Displaced workers are defined as individuals who experienced involuntary separation from their main job with at least 12 months of tenure. Section 4.2 examines the sensitivity of the results to minimum tenure requirements of 24 and 36 months. Involuntary separations include plant or company closures or moves, slack work or business conditions, position or shift abolishment, and other involuntary reasons.[12] Appendix Table A1 reports

[10]The sample further excludes individuals with missing values on analysis variables and residents of Puerto Rico, U.S. territories, or foreign countries. Appendix Figure A2 reports the sample-selection counts.

[11]Never-displaced controls are selected on a favorable future, so comparisons against them can mix the effect of displacement with differences in later job stability. Krolikowski (2018) discusses this problem and proposes a related, though different, approach to choosing the comparison group.

[12]The selected separation reasons follow those used by Kuka and Stuart (2025) for the 2014 SIPP panel, except that I exclude the completion of temporary or seasonal jobs. The resulting set also matches that used by Schaller and Zerpa (2026). These categories apply to the 2014 and 2018 panels used here.

how treated individuals are distributed across these separation reasons. When a worker holds multiple jobs in a given month, the main job is identified based on weekly hours worked, job tenure, and monthly earnings. Although SIPP records job spells at the weekly level, I date each displacement to the calendar month in which the job ended.

Health insurance coverage measures Health insurance coverage is constructed at the individual-month level from survey responses on the type of coverage and on who holds the policy. Appendix Table A2 summarizes the full classification.

Coverage is first classified by source. Private coverage is obtained through a current or former employer, purchased directly from an insurer, or obtained through a union, a professional association, or a school. Public coverage consists of Medicaid, Medicare, military coverage, and other public coverage. Because the sample excludes anyone ever covered by Medicare, military, or other public coverage, public coverage within the sample is Medicaid.

Private coverage is classified along a second dimension, namely who holds the policy. SIPP identifies whether the policyholder is the respondent, the respondent's spouse, or another person. This distinction does not apply to Medicaid.

Combining the two dimensions, I use four monthly coverage indicators as the main outcomes: (1) own employer-sponsored coverage, defined as coverage from a current or former employer with the respondent as policyholder; (2) own directly purchased coverage, defined as coverage purchased directly from an insurer with the respondent as policyholder; (3) spouse-held private coverage, defined as private coverage from any source with the spouse as policyholder; and (4) Medicaid.[13]

In addition to these four indicators, I examine two supplementary outcomes that capture the impact of displacement on net coverage: any insurance and being uninsured. Any

[13]The two own-name indicators are restricted by source, while spouse-held coverage is not. Own-name private coverage obtained through a union, a professional association, or a school is therefore captured by neither own-name indicator, although it enters the any-insurance indicator. Such coverage accounts for 1.6 percent of individual-months in the estimation sample (232 of 14,689, unweighted).

insurance equals one if the individual holds private coverage from any source, regardless of whether the policyholder is the respondent, the spouse, or another person, or is covered by Medicaid; being uninsured is its complement. Within the estimation sample, these two categories are exhaustive, because the sample restrictions remove Medicare, military, and other public coverage.

Three points deserve emphasis here. First, a displaced worker who remains on COBRA continuation coverage is counted as holding own employer-sponsored coverage, since such coverage is expected to be reported as coverage from a former employer. Appendix Figure A5 shows how own employer-sponsored coverage splits between current-employer and former-employer sources around displacement. Second, coverage is recorded at the monthly level rather than at a point in time. Section 4 discusses how this affects the interpretation of the estimate for the month of displacement. Third, these indicators are not mutually exclusive, because an individual can hold more than one type of coverage in the same month.

2.3 Descriptive analysis

Before moving to estimation, this section provides descriptive analysis based on the sample defined above. Panels A and B of Appendix Figure A3 show the distribution of displacement timing for the SIPP 2014 and 2018 panels, respectively, for the 852 individuals in the analysis sample. This includes workers displaced during the estimation window and those displaced after it who serve as controls within the window. The distribution reveals considerable variation in the timing of displacement, which provides the identifying variation for the estimation strategy discussed in Section 3.

Panels A and B of Figure 1 plot average health insurance coverage rates over time for different displacement cohorts in the SIPP 2014 and 2018 panels, respectively. Each cohort is defined by the calendar month of displacement, with cohorts aggregated into four-month groups for presentation. Calendar time is also grouped into four-month

intervals. In both panels, own ESHI coverage declines around displacement for most cohorts. Post-displacement recovery patterns vary across cohorts, with some showing recovery in own ESHI coverage and others exhibiting more persistent coverage losses. The never-treated group consists of workers displaced in 2016 in the 2014 panel and in 2019 in the 2018 panel. Own ESHI coverage in this group remains relatively stable throughout the respective estimation periods of 2014–2015 and 2017–2018.

Table 1 presents baseline demographic characteristics across displacement cohorts, grouped by half-year of displacement, and their corresponding control groups, separately for the 2014 and 2018 SIPP panels. Survey-weighted normalized differences are computed between treated and control samples within each aggregated cohort and panel. Following Baker et al. (2025), I use the rule of thumb from Imbens and Rubin (2015) that absolute normalized differences above 0.25 indicate potentially problematic imbalance. Most comparisons fall below this threshold. In the 2014 panel, differences exceeding this threshold occur for marital status, spousal employment, spousal full-time employment, and Medicaid expansion state residence. In the 2018 panel, they occur for earnings, the household income-to-poverty ratio, spousal full-time employment, sex, marital status, self-reported health, and daily prescription use. Such baseline imbalances can violate parallel trends if the imbalanced covariates are associated with differential outcome trends absent displacement (Baker et al. (2025)).

Appendix Table A3 reports changes in these characteristics between one month before and one month after displacement, separately for the 2014 and 2018 SIPP panels. Changes in employment-related variables show large normalized differences, which is expected given that the treatment itself is job displacement. Among the other covariates, absolute normalized differences exceed 0.25 only for age in one cohort of the 2018 panel. I control for covariates with baseline imbalances and include age as a time-varying covariate in the 2018 panel, as discussed in Section 3. Changes in covariates likely affected by treatment are not used to select adjustment variables.

Insurance state transitions To provide an overview of how insurance coverage changes around job displacement, I construct mutually exclusive insurance states and track their distribution by event time. For this descriptive exercise, each individual is assigned to one state following a priority ordering: own ESHI from a current employer, own ESHI from a former employer, own direct purchase, spouse's private coverage, Medicaid, other private coverage, and uninsured. Other private coverage comprises private insurance held by workers with none of the preceding sources. Uninsured means no private insurance or Medicaid.[14]

Figure 2 presents the distribution of these insurance states by event time. Panel A pools the 2014 and 2018 SIPP panels, while Panels B and C report them separately. Before job loss, own ESHI accounts for the largest share. The own ESHI share drops sharply at $k = 1$, the first full month after displacement. The uninsured share peaks at 27.44% in that month and falls to 13.58% at $k = 12$, while the own ESHI share partially recovers. These changes describe the distribution of insurance states among workers observed at each event time, whose composition varies over the event window.[15]

3 Empirical strategy

To examine the impact of involuntary job loss on health insurance coverage, I adopt a difference-in-differences (DID) design with staggered treatment timing. This design compares changes in health insurance status around job displacement between treated individuals and those not yet treated or never treated within the sample period. The specific estimator is described below. The notation used in this section follows Gardner et al. (2024)

[14]This ordering is for the descriptive analysis only. The main regression analysis uses non-exclusive coverage indicators, allowing individuals to hold multiple types of coverage simultaneously, and does not separate current from former employer coverage. The split between current and former employer is shown here because continuation coverage through COBRA is expected to be reported as former-employer coverage, though the survey instrument does not identify COBRA directly.

[15]Appendix Table A4 tracks insurance state transitions specifically for individuals who held own ESHI at $k = -1$.

and Roth et al. (2023).

Causal parameter of interest Consider balanced panel data $\{Y_{it}, X_{it}, D_{it}\}$, where $i \in \{1, 2, \ldots, N\}$ indexes individuals and $t \in \{1, 2, \ldots, T = 24\}$ indexes months within each SIPP panel. In the 2014 panel, $t = 1$ corresponds to January 2014 and $t = 24$ to December 2015. In the 2018 panel, the corresponding months are January 2017 and December 2018. Individuals may experience treatment (i.e., job displacement) at different points between February 2014 and December 2015 in the 2014 panel, or between February 2017 and December 2018 in the 2018 panel. Treatment is regarded as an absorbing state. Let D_{it} indicate whether individual i has been treated by time t, and G_i denote the month of job displacement for individual i, that is, $G_i = \min\{t : D_{it} = 1\}$. Individuals are partitioned into cohorts $g \in \{2, 3, \ldots, 24, \infty\}$, where cohort g consists of individuals displaced at time g. The cohort $g = \infty$ represents "never-treated" individuals. In this analysis, this group consists of those displaced outside their panel's estimation window, during 2016 (wave 4) in the 2014 panel or during 2019 (wave 3) in the 2018 panel. For each individual i, let $Y_{it}(g)$ denote the potential outcome at time t if treated at time g. Then $Y_{it}(\infty)$ represents the potential outcome had the individual never been treated during the sample period, $t = 1, ..., 24$.

Let $p(i) \in \{2014, 2018\}$ denote individual i's SIPP panel. Define the cohort-time average treatment effect on the treated in panel p as $ATT_p(g, t) = \mathbb{E}[Y_{it}(g) - Y_{it}(\infty) \mid G_i = g, p(i) = p]$. This measures the average treatment effect at time t for treated individuals in cohort g of panel p. The causal parameters of interest are duration-specific average treatment effects on the treated (ATTs), also known as event-study parameters, which capture how health insurance coverage evolves relative to the timing of displacement.

To construct the aggregation weights, I start with each individual's longitudinal survey

weight w_i and normalize it as

$$\tilde{w}_i = \frac{w_i}{\bar{w}_{p(i)}},$$

where $\bar{w}_p$ is the mean survey weight across all individuals in panel p's analysis sample, including never-treated individuals, before restricting observations to the event-study window. This preserves relative survey weights within each panel.

Let k index time relative to displacement, and let $\mathcal{I}_{pg}(k)$ contain individuals in treated cohort g of panel p who are observed at time $g + k$ within the analysis window. Summing their normalized survey weights gives the cohort total,

$$W_{pg}(k) = \sum_{i \in \mathcal{I}_{pg}(k)} \tilde{w}_i.$$

The corresponding panel total is $W_p(k) = \sum_{g \in \mathcal{G}_p(k)} W_{pg}(k)$, where $\mathcal{G}_p(k)$ contains the treated cohorts observed at event time k in panel p.

These totals determine the aggregation weights,

$$w_{pg}(k) = \frac{W_{pg}(k)}{W_p(k)},$$
$$\omega_p(k) = \frac{W_p(k)}{\sum_{q \in \{2014,2018\}} W_q(k)}.$$

Here, $w_{pg}(k)$ is the survey-weighted relative frequency of cohort g among treated individuals observed at event time k in panel p, and $\omega_p(k)$ is panel p's share of the normalized survey weights across both panels at that event time.

The panel-specific and pooled event-study parameters are then

$$ATT_p^{ES}(k) = \sum_{g \in \mathcal{G}_p(k)} w_{pg}(k) ATT_p(g, g+k),$$
$$WATT_k^{ES} = \sum_{p \in \{2014,2018\}} \omega_p(k) ATT_p^{ES}(k).$$

Both the contributing cohorts and the panels' shares can vary with k. Negative values of k correspond to pre-treatment periods, while positive values correspond to post-treatment periods.

The sample construction matters for the interpretation of these parameters. The sample includes only workers who are displaced during or shortly after the estimation window. Untreated observations therefore come from workers observed before their own displacement, not from workers who stay employed throughout the panel (see Section 2). As a result, $WATT_k^{ES}$ measures the effect of earlier displacement relative to later displacement within a population already on a path toward separation, rather than the effect of job loss relative to stable employment. To check how much the results depend on this choice of the comparison group, Section 4.2 re-estimates the model with two alternative comparison groups: one adds workers who are never displaced, and the other excludes workers displaced after the estimation window and uses only pre-displacement observations from workers displaced within that window as controls.

Identifying assumptions Identification of these causal parameters relies on three key assumptions following the recent literature on staggered DID designs (Roth et al. (2023); Baker et al. (2025); Gardner et al. (2024)): (i) The conditional parallel trends assumption requires that, within each panel p, for all $t \neq t'$, $g \neq g'$, and each covariate value X_i, $\mathbb{E}[Y_{it}(\infty) - Y_{it'}(\infty) \mid G_i = g, X_i, p(i) = p] = \mathbb{E}[Y_{it}(\infty) - Y_{it'}(\infty) \mid G_i = g', X_i, p(i) = p]$. This assumes that, conditional on covariates, both treated and untreated individuals within the same panel would have experienced parallel trends in potential outcomes absent

treatment. (ii) The no anticipation assumption states that $Y_{it}(g) = Y_{it}(\infty)$ for all i and $t < g$, meaning individuals do not alter their behavior in anticipation of future job displacement. (iii) The overlap condition requires that for all i, there exists some t where $D_{it} = 0$, that is, every individual has at least one untreated period and that each relative-time bin k contains a positive share of treated observations.

Estimation procedure For estimation, the two-step DID approach developed by Gardner et al. (2024) is adopted. This estimator follows an imputation approach closely related to that of Borusyak et al. (2024) and provides a convenient way to compute standard errors through its GMM framework.

In the first stage, I estimate the model for untreated potential outcomes specified as equation (1), using only untreated observations (i.e., $D_{it} = 0$).

$$Y_{it}(\infty) = \lambda_i + \alpha_{p(i),t} + X_{it}^{'}\gamma_{p(i)} + u_{it} \tag{1}$$

Here, λ_i represents individual fixed effects, $\alpha_{p(i),t}$ captures panel-specific time fixed effects, and X_{it} contains baseline characteristics interacted with time indicators and any time-varying covariates included in the specification. The coefficient vector $\gamma_{p(i)}$ is panel-specific. Following Baker et al. (2025), covariates are selected based on the baseline balance table: a covariate is included if its absolute normalized difference exceeds 0.25 between treated and control samples for any cohort comparison. Because the two panels cover disjoint calendar periods and disjoint samples, this diagnostic speaks to parallel trends within each panel separately, so covariates are selected panel by panel and all coefficients are panel-specific.[16] In the second stage, I define $W_{k_{it}} = \mathbb{1}[t - G_i = k]$ as an indicator for whether

[16]For the 2014 panel the selected covariates are married, spouse employed (conditional on married), spouse full-time (conditional on married and spouse employed), and Medicaid expansion state indicator. For the 2018 panel they are employed, earnings (conditional on employed), married, spouse employed, spouse full-time, female, good self-reported health, daily prescription use, and household income-to-poverty ratio. Employment and spousal employment indicators are also included because the selected earnings and spousal full-time employment variables are conditional on these statuses. Baseline characteristics are measured at January 2014 for the 2014 panel and January 2017 for the 2018 panel. Age enters the 2018

individual i is k months away from treatment at time t, and regress the adjusted outcome $Y_{it} - (\hat{\lambda}_i + \hat{\alpha}_{p(i),t} + X_{it}^{'}\hat{\gamma}_{p(i)})$ on the set of relative-time indicators $\{W_{k_{it}}\}$.

Heterogeneity analysis The heterogeneity analysis examines how displacement effects vary by subgroup using an interaction specification. I consider two dimensions: workers' household income-to-poverty ratio and state Medicaid expansion status, both measured at the month prior to displacement ($t = g - 1$). Specifically, the second-stage regression interacts the event-time indicators $\{W_{k_{it}}\}$ with a subgroup dummy S_i.

4 Estimation results

This section presents the empirical findings on how job displacement affects health insurance coverage. Section 4.1 examines the main effects on own ESHI, own DPHI, spouse's private coverage, and Medicaid, together with the net effect on any coverage and on uninsurance. Section 4.2 assesses the sensitivity of these findings. Section 4.3 then explores heterogeneity along two dimensions: workers' income before displacement and state Medicaid expansion status. Section 4.4 examines the robustness of the expansion contrast and explores Medicaid entry and exit. All results use the covariate-adjusted specification described in Section 3.

4.1 Main analysis

Table 2 and Figure 3 present the event study estimates for the four insurance-source outcomes and uninsurance. The table also reports estimates for any health insurance coverage. Any coverage is the union of private coverage and Medicaid in the analysis sample, and uninsurance is its complement. SIPP records health insurance coverage as a spell with a beginning and ending month, so a month is recorded as covered if the spell

panel as a time-varying covariate. Variables that displacement itself can move are treated as mechanisms and are not used for adjustment.

overlapped that month, not only if coverage lasted the entire month. Because $k = 0$ is the month in which displacement occurred, the worker was still employed during part of that month. The $k = 0$ coefficient therefore mixes pre- and post-displacement exposure, and $k = 1$ is the first event-time coefficient that reflects a full month of post-displacement status. Pre-displacement coefficients ($k = -6$ through $k = -1$) are small and statistically insignificant for the four insurance-source outcomes, while those for any coverage and uninsurance are significant at the 5 percent level at $k = -1$.

Own-name ESHI coverage falls sharply after displacement. The effect at $k = 1$ is a 11.7 percentage point decline and is still 10.1 percentage points at $k = 3$. The deficit then narrows. By $k = 12$ it is 3.2 percentage points and no longer statistically significant, indicating some recovery within the first year.

The two rightmost columns of Table 2 report the net effect on coverage. Uninsurance rises by 7.0 percentage points at $k = 1$, significant at the 1 percent level. The gap then narrows. At $k = 3$, the increase is 3.4 percentage points and is no longer statistically significant. From $k = 6$ onward, the point estimate is negative. By $k = 12$, any coverage is 8.2 percentage points higher than in the absence of displacement.

Medicaid coverage increases following displacement, with a 1.8 percentage point rise at $k = 1$ that grows to 3.4 percentage points by $k = 6$. The Medicaid response remains positive and statistically significant through the end of the event window, reaching 5.9 percentage points at $k = 12$.

Own DPHI and spouse's private coverage show negligible responses throughout the post-displacement period, and none of the coefficients are statistically significant at conventional levels.[17] These estimates suggest that own DPHI and spouse's private coverage play a limited role, on average, in offsetting the loss of own ESHI after displacement. I

[17]Figure 2 shows insurance-state shares among displaced workers, whereas the regressions estimate coverage changes relative to the comparison group. The figure assigns each worker to a single category according to the ordering in Section 2.3. A worker with both own ESHI and spousal coverage can move into the spousal category when own ESHI ends, even if spousal coverage itself is unchanged. The regressions allow multiple sources of coverage, so the descriptive shares and estimated effects need not move together.

discuss possible explanations for this limited response of own DPHI in Section 6.

4.2 Sensitivity of the findings

Pre-trend testing The key identifying assumption is parallel trends. To assess its validity, I implement two complementary testing procedures. First, following Borusyak et al. (2024), I estimate an augmented first-stage model using only untreated observations that includes leads of treatment status. I then conduct a joint Wald test of the null hypothesis that all pre-treatment coefficients equal zero. Second, I implement the pre-trend test proposed by Gardner et al. (2024), which operates within the two-stage framework. The former test examines whether future treatment status predicts current outcomes among not-yet-treated observations. The latter test uses a modified first-stage sample that restricts eventually-treated units to the period immediately before displacement, so that, under the normalization that parallel trends holds at that period, the pre-treatment coefficients measure average deviations of eventually-treated units (those displaced during the sample period) from never-treated units' trends (those displaced outside the sample window). Appendix Table A6, Panels A and B, reports the Gardner and Borusyak et al. tests. Both fail to reject the null hypothesis of zero pre-treatment effects for the four insurance-source outcomes.

Although neither test rejects the null, their power to detect economically meaningful pre-trends may be limited. I therefore use the sensitivity analysis of Rambachan and Roth (2023) to assess how robust the findings are to violations of parallel trends. I allow the untreated outcome gap between treated and control groups to change after displacement, bounding each monthly change by $\bar{M}$ times the largest absolute monthly change before displacement. Larger values of $\bar{M}$ allow larger violations of parallel trends.

Appendix Table A7 reports the results for own ESHI at $k = 3$ and $k = 12$ and Medicaid at $k = 12$, with estimates recentered at $k = -1$. The robust 95 percent confidence interval for own ESHI at $k = 3$ first includes zero at approximately $\bar{M} = 0.88$, compared with

$\bar{M} = 0.12$ for Medicaid at $k = 12$. The interval for own ESHI at $k = 12$ already includes zero at $\bar{M} = 0$. The month-twelve Medicaid increase is therefore sensitive to departures from parallel trends that are small relative to pre-displacement trend differences. Under the relative-magnitude restriction, monthly departures can accumulate over time, so longer-horizon effects allow a wider range of possible bias.

Alternative comparison groups The control group in the main specification consists entirely of eventual displacees. Control observations come from workers displaced within the estimation window, observed before their own displacement, and from workers displaced in 2016 (wave 4) in the 2014 panel or in 2019 (wave 3) in the 2018 panel. I consider two alternative comparison groups, which change both the composition of the controls and the interpretation of the estimated displacement effects. The first check adds workers who are never displaced and satisfy the remaining sample criteria.[18] This broadens the comparison beyond eventual displacees. The second check excludes workers displaced in wave 4 of the 2014 panel or wave 3 of the 2018 panel and uses only pre-displacement observations from workers displaced within the estimation window as controls.[19] This comparison therefore relies entirely on differences in displacement timing among workers displaced within that window. Appendix Figure A7 and Table A10 report the results for the four insurance-source outcomes.

Both alternatives preserve the decline in own ESHI and the small response of own DPHI, although the ESHI decline is somewhat smaller when controls are restricted to workers displaced within the estimation window. The Medicaid response is smaller under both alternatives, particularly at longer horizons. At $k = 12$, the estimated increase is 5.9

[18] "Never displaced" means that the worker experiences no displacement as defined in Section 2 during the observation period. These workers may still leave jobs in ways that do not meet this definition, for example through voluntary quits or separations from jobs held for less than twelve months. The expanded sample contains 19,036 individuals, including 18,000 never-displaced workers, and also restores workers displaced during 2020 in the 2018 panel. Estimation for that panel still uses only 2017–2018 observations.

[19] The final sample month of each panel is dropped because no untreated observations remain. Covariates are re-selected within this sample, and survey weights are renormalized to have person-level mean one within each panel.

percentage points in the main specification, compared with 2.7 percentage points when never-displaced workers are added and 1.5 percentage points when workers displaced after the window are excluded. Spouse's private coverage follows a similar pattern to the main results when never-displaced workers are added. When workers displaced after the window are excluded, however, the later estimates become negative, with wide confidence intervals.

The changes in Medicaid and spouse's private coverage may reflect differences in underlying coverage trends across the groups used as controls. The main specification compares workers who all eventually experience displacement, while retaining workers displaced after the window to provide untreated comparison observations at longer horizons. Appendix Tables A5 and A11 illustrate the role of these workers. At $k = 12$, pooling the counts across the two panels, excluding them reduces treated observations from 265 to 242, but control observations from 3,887 to 1,302, about one-third of the main sample's control observations. These counts refer to person-month observations in the calendar months contributing to that event time. The main specification thus retains a comparison within eventual displacees and a larger pool of untreated observations at longer horizons, while the alternative specifications show that the estimated Medicaid and spouse-coverage responses depend on the choice of comparison group.

Minimum tenure requirement The displacement definition in Section 2 requires at least twelve months of tenure on the lost job, as in Flaaen et al. (2019) and East and Simon (2024).[20] Displacements from jobs held for longer periods may be more plausibly exogenous and entail larger losses. To assess sensitivity to the minimum tenure requirement, I raise the threshold to twenty-four and thirty-six months, rebuild the sample under each threshold, and re-estimate the covariate-adjusted event study for the four outcomes. The

[20]Tenure is calculated as the number of months between the job's start month and the month of displacement, using start-date information across waves. When the information needed to identify the start month is missing, I apply the following conservative rules. If only the start year is known, I use December of that year. If the start date remains unidentified and the job is first observed in January of its first observed year, I use that January. Separations for which tenure remains unknown are not classified as eligible displacements.

covariate set is held fixed at the main selection. The higher thresholds retain 446 and 356 of the 617 displaced workers in the main sample (72 and 58 percent). Appendix Figure A4 plots the three event-study paths for each outcome, and Appendix Table A8 reports the estimates.

Own ESHI, own DPHI, and spouse's private coverage show broadly similar patterns across the three minimum tenure requirements. The decline in own ESHI at $k = 1$ is 11.7, 11.3, and 11.2 percentage points under the twelve-, twenty-four-, and thirty-six-month requirements. Although some point estimates differ, these three outcomes do not show a consistent ordering by the minimum tenure requirement throughout the post-displacement period. The Medicaid response is smaller, particularly under the thirty-six-month requirement, but increases at longer horizons under all three minimum tenure requirements. At $k = 12$, the estimates are 5.9, 4.8, and 4.4 percentage points. These results suggest that the main qualitative patterns are not driven by the inclusion of workers with only one to two years of tenure.

Former-employer coverage The main definition of own ESHI includes coverage from both current and former employers. COBRA continuation coverage is expected to be reported as former-employer coverage, although SIPP does not identify COBRA directly. Appendix Figure A5 shows that former-employer coverage accounts for a small share of displaced workers and does not rise markedly after displacement. To assess whether its inclusion affects the results, I re-estimate the model using only current-employer coverage, with the same sample and covariate adjustment as in the main specification. Appendix Figure A6 and Table A9 compare the two definitions. The initial ESHI decline is nearly identical, at 11.7 percentage points when former-employer coverage is included and 11.4 percentage points when it is excluded at $k = 1$. While the difference between the estimates becomes more noticeable as the coverage deficit narrows, both definitions show a sharp decline followed by a narrowing coverage deficit. Thus, the main pattern does not depend

on including former-employer coverage.

4.3 Heterogeneity analysis

Heterogeneity by pre-displacement income Workers differ in the income their household has when ESHI ends. Private coverage requires paying a premium out of pocket, so demand for it may differ across the income distribution. To examine these differences, event-time indicators are interacted with an income group dummy, stratifying by the household income-to-poverty ratio measured as of the month before displacement ($g-1$). This ratio is total household income divided by the poverty threshold for that household, so it counts the income of other household members and is not limited to the worker's own labor income. The cut is at 400% of FPL, the upper limit of the ACA premium subsidy range during the study period. Income before job loss is related to eligibility for Medicaid and for marketplace subsidies, but eligibility depends on income after displacement. This split therefore does not identify which workers are eligible.[21] Figure 4 presents the results. Appendix Table A12, Panel A, reports selected estimates underlying the figure. Panel B reports treated-worker and control-observation counts at each event time.

Own ESHI coverage falls sharply in both income groups immediately after displacement. The initial decline is somewhat larger among lower-income workers, but their coverage gap subsequently closes, while the decline persists among higher-income workers. Medicaid enrollment increases more among lower-income workers, reaching 7.5 percentage points at twelve months compared with 3.6 points among higher-income workers. These patterns suggest that Medicaid plays a relatively more important role as a safety net following job loss among lower-income workers.

[21] Eligibility for Marketplace premium tax credits generally depends on projected annual household income. MAGI-based Medicaid eligibility for new applicants is based on current monthly household income. The ratio used here is a single month of income before job loss. Displacement is itself an income shock. Some workers above the cut at $g-1$ become subsidy-eligible after job loss, and some below it become eligible for Medicaid, which rules out premium subsidies. In non-expansion states, workers whose income falls below 100% of FPL generally qualify for neither Medicaid nor Marketplace premium tax credits unless they qualify for Medicaid under traditional eligibility categories.

DPHI responses are small in both income groups. If demand for private coverage depended on the resources described above, higher-income workers would be expected to respond more, since they have more resources to draw on. The point estimates do not show this pattern. The DPHI response is no larger among higher-income workers.

Heterogeneity by Medicaid expansion status To examine how post-displacement insurance dynamics differ by Medicaid expansion status, event-time indicators are interacted with a Medicaid expansion status dummy in the second stage, where expansion status is measured as of the month before displacement ($g-1$).[22] I report the estimates separately for the 2014 and 2018 SIPP panels to distinguish responses during 2014–2015 from those during 2017–2018. Panels A and B of Figure 5 present the results for the SIPP 2014 and 2018 panels, respectively.[23] The classification is fixed at displacement, so workers in states that expand afterward remain in the non-expansion group even when Medicaid expansion becomes available during their follow-up period.[24] Appendix Table A13 reports selected group-specific estimates and treated-worker support for the later Medicaid estimates.

In the 2014 SIPP panel, the initial Medicaid point estimates are positive in both state groups, although those in non-expansion states are statistically insignificant. The paths subsequently diverge. In expansion states, the increase grows from 4.7 percentage points at $k = 3$ to 10.1 percentage points at $k = 12$. In non-expansion states, the point estimates are positive through $k = 11$, but the increase fades toward the end of the event window and is close to zero at $k = 12$. Own ESHI coverage falls sharply in both groups. At $k = 1$, own ESHI declines by 12.4 percentage points in expansion states and 14.5 percentage points in non-expansion states. The deficit narrows in non-expansion states relative to expansion states at later horizons. Own DPHI and spouse's private coverage show small initial

[22] In both panels, workers are grouped by Medicaid expansion status in the month before displacement ($g-1$). For first-stage covariate adjustment, the 2014 panel uses an expansion indicator measured in January 2014, whereas the 2018 panel includes no expansion indicator as a covariate.

[23] Pooled estimates are reported in Appendix Figure A8.

[24] The expansion group also includes states that expanded before 2014. Section 4.4 examines sensitivity to excluding transitioning states and early expanders.

responses. Their point estimates rise at longer horizons in expansion states, although these later estimates are imprecise.

In the 2018 SIPP panel, Medicaid coverage in expansion states shows no sustained increase comparable to that in the 2014 panel. In non-expansion states, however, the point estimates rise during the middle of the event window, reaching 15.5 percentage points at $k = 9$ and remaining at 11.4 percentage points at $k = 12$. These later estimates are imprecise, with only 26 treated workers contributing at $k = 12$ (Appendix Table A13, Panel B). Own ESHI declines initially in both groups. In non-expansion states, the point estimates return to near zero shortly after displacement and fluctuate thereafter. In expansion states, coverage recovers steadily over the event window. Own DPHI responses are small. The later point estimates for spouse's private coverage are negative in expansion states and positive in non-expansion states, with wide confidence intervals. I return to the absence of a sustained Medicaid increase in expansion states in Section 4.4, where I examine post-displacement income qualification and Medicaid entry and exit.

4.4 The expansion contrast: robustness and decomposition

State-specific trends and early expanders To allow for potential state-specific trends in untreated outcomes, such as those arising from changes in enrollment outreach or local labor market conditions, I add state-by-calendar-month fixed effects to the first-stage regression. I then exclude states that expanded Medicaid during the estimation window and, in a further specification, also exclude states that had expanded coverage before 2014.[25] Panels A–D of Appendix Figure A9 report the results separately for each panel and expansion group. Table A14 reports estimates and confidence intervals at selected

[25]The transitioning states are Michigan, New Hampshire, Pennsylvania, Indiana, and Alaska in the 2014 panel. No states expanded during the 2017–2018 estimation window of the 2018 panel. The additional exclusions cover early-expansion states (California, Connecticut, the District of Columbia, Minnesota, New Jersey, and Washington) and states with broader pre-ACA coverage (Delaware, Massachusetts, New York, and Vermont), following KFF (2012) and Miller et al. (2021). Both exclusion specifications retain state-by-calendar-month fixed effects, and the final specification excludes both sets of states.

later horizons.

In the 2014 panel, allowing for these trends preserves the qualitative Medicaid patterns and does not attenuate the increase in expansion states. The increase becomes larger with state-by-calendar-month fixed effects and changes little when transitioning states are excluded. Further excluding states with pre-2014 coverage expansions reduces the Medicaid increase from 15.2 to 9.2 percentage points at $k = 12$, but the pattern of rising coverage at longer horizons remains. In non-expansion states, the pattern of an initial increase that fades toward the end of the event window remains broadly similar across specifications. In the 2018 panel, the Medicaid patterns remain broadly similar across specifications in both expansion and non-expansion states.

State-level clustering To assess whether inference for the Medicaid estimates reported by expansion status is sensitive to dependence across workers within states, I compare individual- and state-cluster inference in Table A14. Within each specification, changing the clustering level leaves the sample and point estimates unchanged. The later Medicaid increase in the 2014 expansion group remains evident under conventional state-cluster inference, including after adding state-by-calendar-month fixed effects and applying the cumulative state exclusions. At $k = 12$, the state-cluster 95% confidence intervals exclude zero in all four specifications. The estimate is 14.75 percentage points with state-by-calendar-month fixed effects (95% CI [4.13, 25.38]) and 9.22 percentage points after both sets of state exclusions (95% CI [1.93, 16.51]). In the 2018 expansion group, the estimates show no comparable sustained increase, and the later confidence intervals remain wide.

Conventional state-cluster inference may nevertheless overstate precision when few states contribute treated observations (MacKinnon et al., 2023). I therefore also use a state-level wild-cluster bootstrap for the state-by-calendar-month fixed-effects specification.[26] At the selected horizons, WCU confidence intervals are wider than conventional state-

[26] I use the unrestricted wild-cluster bootstrap (WCU), which allows pointwise inference for all event-time coefficients using the same bootstrap draws, reducing the computational burden.

cluster intervals. For the 2014 expansion group, the WCU interval excludes zero only at $k = 9$, and includes zero at $k = 12$ while the conventional state-cluster interval excludes zero. The 2018 expansion estimates also remain indistinguishable from zero.

Sensitivity to parallel trends To assess the sensitivity of this increase to departures from parallel trends, I apply HonestDiD to the baseline estimates for the 2014 expansion group (Table A7, Panel B). The month-12 estimate relative to month −1 is 9.57 percentage points (ordinary 95% CI [2.77, 16.36]).[27] The robust 95% confidence interval first includes zero at approximately $\bar{M} = 0.32$.

Pre-ACA Medicaid responses in later-expanding states Using the monthly Medicaid information available in the 2008 SIPP panel, I compare pre-ACA responses to displacement over 2011–2012 between states that implemented Medicaid expansion by December 2019 (later-expander states) and other states. This comparison examines whether the larger Medicaid responses observed in expansion states in the 2014 SIPP panel were already present before the ACA. Medicaid coverage rises after displacement in both groups, with broadly similar paths and no clear pattern of larger increases in states that later expanded Medicaid (Figure A10). The pre-ACA data and sample construction are described in Section 5, which also presents the wave-level comparison with the post-ACA period.

Income qualification, entry, and exit To examine whether the lack of a sustained Medicaid increase in the 2018 expansion group reflects family incomes remaining above the income threshold after displacement, I use an indicator for monthly family income at or below 138% of the poverty threshold as a proxy for Medicaid eligibility. I estimate the effect of displacement on this indicator using the same specification as for Medicaid coverage.[28] Panels A and B of Figure A11 report the results for the 2014 and 2018 SIPP

[27] Before applying HonestDiD, I subtract the estimated coefficient at $k = -1$ from each event-time coefficient and transform the covariance matrix accordingly.

[28] The proxy uses the SIPP family income-to-poverty ratio and captures only the income requirement, rather than a complete determination of Medicaid eligibility.

panels, respectively. In both panels, displacement initially raises the probability of meeting the income threshold, with the increase diminishing over time. In the 2018 expansion group, income qualification also rises after displacement. The flat Medicaid coverage response in this group therefore cannot be explained simply by a lack of increased income qualification.

I next examine whether the lack of a sustained Medicaid increase in the 2018 expansion group could reflect increased entry being offset by increased exit. I separate workers by their Medicaid enrollment in the month before displacement. I measure both observed and counterfactual changes from this baseline. I use two-stage DID to estimate the effect of displacement on entry among initially non-enrolled workers and on exit among initially enrolled workers. I weight these effects by the shares of initially non-enrolled and enrolled workers to obtain their contributions to the overall coverage response. The net effect is entry minus exit. Figure A12 and Table A15 report the results.

In the 2014 panel, the decomposition point estimates show a sustained increase in entry in expansion states, while the smaller initial increase in non-expansion states fades. In the 2018 panel, entry increases in both groups, with larger point estimates in non-expansion states over most of the post-displacement period. At longer horizons, the estimated reduction in exit further raises the net coverage response in the 2014 expansion group. In the 2018 expansion group, the estimated contribution of increased exit exceeds that of increased entry in most post-displacement months, leaving the net response small or negative. Non-expansion states show no comparable sustained increase in exit. These patterns suggest that increased exit is one possible explanation for the lack of a sustained Medicaid increase in the 2018 expansion group.

5 Comparison of displacement effects between pre-ACA and post-ACA periods

In this section, I compare the effects of job displacement on health insurance coverage before and after the ACA. The comparison examines whether insurance losses became smaller and how responses across coverage sources changed. It describes differences in displacement effects across periods but does not aim to isolate the causal effect of the ACA itself.

Data and comparison design For the pre-ACA period, I use the 2008 SIPP panel to study displacement effects in 2011–2012, shortly before the ACA's major coverage expansions in 2014.[29] Following the sample design used for the post-ACA analysis, I include workers displaced during 2011–2012 and those displaced in 2013, but restrict estimation to observations from 2011–2012. Workers displaced in 2013 therefore contribute to the comparison group throughout the estimation window. For the post-ACA period, I use the pooled 2014 and 2018 panels described above. Information on the source and policyholder of private insurance in the 2008 panel is recorded at the wave level. [30] I therefore estimate coverage responses at the wave level in the 2008 SIPP panel. For a comparable analysis of the post-ACA panels, I group monthly observations into four-month periods and measure whether workers held each type of coverage at any time during each period.[31]

I estimate the two-stage DID model at the wave level, excluding workers displaced in their first observed wave so that each displaced worker has at least one pre-displacement wave.[32] Event time $k' = 0$ denotes the wave containing displacement. I recenter each

[29] In the 2008 panel, the selected separation reasons are being on layoff, employer bankruptcy, the employer selling the business, and slack work or business conditions. These reasons match those used by Flaaen et al. (2019), Kuka (2020), and Schaller and Zerpa (2026).

[30] Medicaid coverage is observed monthly, as used in the preceding analysis of later-expander states.

[31] Coverage from each source indicates enrollment at any time during the observed months of the wave or interval. Uninsurance indicates an absence of coverage throughout those months. The SIPP panels differ in interview frequency and recall periods. These differences could contribute to differences in estimated coverage paths through reporting errors, including seam bias, even after aggregation to comparable intervals.

[32] Relative to the samples in Appendix Figure A2, this restriction excludes 33 pre-ACA workers and

period's estimates by subtracting the coefficient for the preceding wave, $k' = -1$, so that the comparison measures changes from a common pre-displacement reference point. Figure 6 reports the comparison. In the pre-ACA sample, the pre-displacement own ESHI coefficients show no sustained downward pattern, and their 95% confidence intervals include zero. Table A16 reports the estimates in Panel A and treated-worker support in Panel B.

Main results Figure 6 shows that the clearest difference across insurance sources is in own ESHI, which falls more sharply before the ACA. At the wave following displacement ($k' = 1$), own ESHI declines by 30.6 percentage points in the pre-ACA period, compared with 8.8 points in the post-ACA period (Table A16). The difference narrows at later horizons. Medicaid responses are similar across periods, and the other private coverage sources show no comparably clear differences. The difference in own ESHI losses is also reflected in the response of uninsurance. At $k' = 1$, uninsurance rises by 17.9 points before the ACA and 1.3 points after it, and the larger pre-ACA increase persists at subsequent horizons.

Adjustment for pre-displacement trends To assess whether potential differences in pre-displacement trends account for the larger ESHI losses estimated before the ACA, I add cohort-specific linear trends to the first stage of the two-stage DID model in both periods. These trends are estimated using only untreated observations and extrapolated beyond displacement, drawing on the detrending approach discussed by Goodman-Bacon (2021).[33] Figure A13 shows that the main comparison remains unchanged. Own ESHI losses and increases in uninsurance remain larger before the ACA. At $k' = 1$, the difference in ESHI losses changes from 21.8 to 26.6 percentage points, while the pre/post gap in the

74 post-ACA workers. A further 98 pre-ACA workers have zero longitudinal survey weight and do not contribute to estimation. The wave-level estimation samples therefore contain 324 pre-ACA workers and 778 post-ACA workers.

[33] See Online Appendix of Goodman-Bacon (2021).

increase in uninsurance changes from 16.6 to 17.2 points (Appendix Table A17).

Differences in labor market conditions One concern with the pre/post comparison is that smaller ESHI losses after the ACA may reflect more favorable labor market conditions and smaller employment losses following displacement. To assess whether ESHI losses are smaller relative to employment losses as well, I compare the two responses across periods, recentering both at the wave before displacement ($k' = -1$) (Figure A14, Panel A). At $k' = 1$, employment declines by 64.3 percentage points before the ACA and 37.7 points after it. ESHI losses are also smaller after the ACA relative to the decline in employment. The ratio of the ESHI decline to the employment decline is 0.475 before the ACA and 0.234 after it, and remains lower in the post-ACA period at subsequent horizons (Panel B). One possible explanation is that workers are more likely to obtain ESHI upon re-employment, or obtain it sooner, in the post-ACA period. These patterns suggest that the adverse consequences of displacement for ESHI coverage may have become less severe, although the comparison does not cleanly isolate this change from differences in economic conditions or worker composition.

Sensitivity to departures from parallel trends I use HonestDiD to assess the sensitivity of the pre-ACA ESHI response and the pre/post differences to potential departures from parallel trends. At $k' = 1$, the robust 95% confidence interval for the pre-ACA ESHI response first includes zero at approximately $\bar{M} = 1.01$ (Table A7, Panel C). For the differences between the pre-ACA and post-ACA displacement effects, the corresponding breakdown values are approximately 1.18 for own ESHI and 0.93 for uninsurance (Panel C).[34]

[34]After recentering each period's estimates at $k' = -1$, I subtract the post-ACA estimate from the pre-ACA estimate at each event-time wave. The pre-displacement path of this difference, from $k' = -5$ through $k' = -1$, provides the benchmark for the sensitivity analysis. Recentering sets the reference-wave difference to zero but preserves changes between adjacent waves. The relative-magnitude restriction allows each post-displacement change in the difference in biases across periods to be at most $\bar{M}$ times its largest absolute change before displacement. I increase $\bar{M}$ to find the value at which the robust interval first includes zero.

6 Discussion

The analyses in Sections 4 and 5 show that job displacement still causes an immediate loss of own ESHI and an increase in uninsurance in the post-ACA period. Medicaid enrollment rises, with responses varying by income, state expansion status, and period, while directly purchased and spousal private coverage show small average responses. Compared with the pre-ACA period, losses of own ESHI and increases in uninsurance are smaller, although the comparison does not identify the causal effect of the ACA.

Schaller and Zerpa (2026), in concurrent work developed independently of this paper, also study insurance coverage following job loss before and after the ACA. The two studies share several findings but differ in the coverage responses they examine.

For the pre/post comparison, Schaller and Zerpa report private insurance coverage without separately estimating responses by source and policyholder. This paper distinguishes these responses, showing that the clearest difference across periods lies in own ESHI losses, with no comparably clear changes in other private coverage responses.[35] Both studies document smaller insurance losses after the ACA and find little evidence that the Medicaid response to job loss became larger across periods. Schaller and Zerpa also highlight the protection provided by higher pre-displacement Medicaid enrollment and assess their SIPP findings using MEPS data.

For the post-ACA analysis, I separately trace own ESHI, directly purchased coverage, spouse-held coverage, and Medicaid, and examine responses by state expansion status in the 2017–2018 SIPP data, beyond the period covered by their SIPP analysis. The sustained Medicaid increase in expansion states in the 2014 panel is not apparent in the 2018 panel, although the latter estimates are imprecise. The entry and exit estimates suggest that increased exit may offset increased entry in the 2018 expansion group. These findings suggest that maintaining Medicaid coverage, alongside enrollment after job loss, matters

[35] The pre-ACA measure includes private coverage held in the name of anyone other than the respondent, whereas the post-ACA measure is limited to spouse-held private coverage. The comparison therefore does not isolate differences specifically in spousal coverage responses.

for its role as a safety net.

The comparison groups also differ. Schaller and Zerpa compare displaced workers with matched workers who experience no involuntary displacement during the sample period, whereas I compare workers displaced at different times.[36]

Limited responses in directly purchased coverage The small response of directly purchased coverage raises the question of why this source does not play a larger role after displacement. One possibility is that Medicaid provides an alternative, reducing demand for private insurance. A small DPHI response therefore need not indicate that Marketplace coverage is ineffective. To examine this possibility, I compare the displacement effects on income qualification, Medicaid coverage, and own DPHI in expansion states, separately for each panel (Figure A15, Panel A). Income at or below 138% of FPL serves as a proxy for Medicaid income qualification. I examine whether the declining effect on income qualification is accompanied by a decline in Medicaid coverage and an increase in DPHI. In the 2014 panel, the Medicaid response continues to rise as the effect on income qualification diminishes. DPHI increases at longer horizons, but this increase is not accompanied by a decline in Medicaid. The 2018 panel also shows no clear substitution pattern. These paths do not show a clear shift from Medicaid toward directly purchased coverage as the effect on income qualification diminishes.

The income paths also show no sustained increase in the probability of being in the 138–400% FPL income range after displacement in expansion states (Figure A15, Panel B). Thus, the decline in the effect on Medicaid income qualification is not accompanied by a clear increase in this intermediate income group, although these income categories do not measure actual subsidy eligibility. Other factors may also limit DPHI enrollment.

[36]The studies also differ in sample restrictions, estimation, and outcome measurement. Schaller and Zerpa require at least one month of unemployment after displacement and estimate matched event studies using two-month event-time bins. I impose no such unemployment-duration restriction and use two-stage DID, with the pre/post comparison estimated at the wave level. The reference periods also differ, with their estimates measured relative to five to six months before displacement and mine relative to the preceding wave.

Premium costs, enrollment procedures, limited awareness of subsidies, and expectations of returning to ESHI through re-employment could each reduce take-up. The small average DPHI response in 2017–2018 also indicates that this pattern is not confined to the initial years of Marketplace operation.

Limitations First, the monthly post-ACA event study extends only to twelve months after displacement. Much of the job displacement literature emphasizes longer-term effects, and prior work has examined displacement's long-run effects on sources of health insurance coverage (Jolly and Phelan, 2017). The design used here cannot establish whether the coverage patterns documented above persist beyond the first post-displacement year. This limitation matters particularly for own DPHI and spouse's private coverage, whose small estimated responses describe adjustment within the first year rather than their longer-term role.

Second, the analysis covers 2014–2015 in the 2014 SIPP panel and 2017–2018 in the 2018 SIPP panel, and does not account for policy changes after these periods. The enhanced premium subsidies under the American Rescue Plan Act and the Inflation Reduction Act, for instance, may have improved marketplace uptake among displaced workers. Whether these changes alter the patterns documented here is a question for future research.

7 Conclusion

This paper examines how job displacement affects health insurance coverage after the ACA and how these effects compare with those before the ACA. The results provide evidence for an overall assessment of insurance protection against displacement in the post-ACA period. Displacement still causes an immediate loss of own ESHI and an increase in uninsurance, although ESHI losses and the initial increase in uninsurance diminish over subsequent months. Medicaid enrollment rises gradually, particularly among lower-income workers,

but its response varies across states and periods. The sustained increase in expansion states in the 2014 panel is not apparent in the 2018 panel. Directly purchased and spousal private coverage show small average responses.

Comparing displacement effects before and after the ACA suggests that their adverse coverage consequences have become less severe, although the comparison does not identify the causal effect of the ACA. The clearest difference across coverage sources is the smaller loss of own ESHI, both in absolute terms and relative to employment losses. This difference is also reflected in smaller increases in uninsurance. Medicaid and other private coverage responses show no comparably clear differences across periods. Medicaid nevertheless provides additional coverage following displacement, particularly among lower-income workers and in expansion states in the 2014 panel.

This improvement leaves an important policy challenge. Alternative coverage sources do not fully prevent the immediate increase in uninsurance following displacement, making continuity of coverage when employment ends a priority. The point estimates from the Medicaid entry and exit decomposition suggest that increased exit may offset increased entry, offering one possible explanation for the absence of a sustained coverage increase in the 2018 expansion group. This finding motivates further research on the factors that shape both enrollment in Medicaid and the retention of coverage following job loss.

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Table 1: Baseline covariate balance by cohort

	SIPP 2014				SIPP 2018			
Covariate	2014/02–06	2014/07–12	2015/01–06	2015/07–12	2017/02–06	2017/07–12	2018/01–06	2018/07–12
Employed	0.190	0.164	0.072	0.176	0.014	0.240	0.175	0.234
Earnings (if employed)	-0.027	-0.081	-0.093	0.071	0.289	-0.326	-0.022	0.071
Hours (if employed)	-0.021	-0.055	-0.059	0.137	0.027	-0.091	0.081	0.235
Full-time (if employed)	-0.159	0.031	0.013	0.076	-0.005	-0.025	-0.088	0.228
Firm size >50 (if employed)	0.025	-0.076	0.058	0.085	0.150	-0.106	-0.178	-0.084
Service sector (if employed)	-0.031	-0.022	0.076	-0.085	-0.070	-0.057	0.153	0.070
Household income-to-poverty ratio	0.129	0.064	-0.172	0.116	0.276	-0.357	-0.169	0.055
Spouse employed (if married)	0.062	0.323	0.051	0.049	-0.024	-0.050	-0.110	0.066
Spouse full-time (if married & spouse employed)	0.065	0.229	-0.272	-0.035	-0.139	0.559	0.081	0.113
Age	-0.061	0.094	-0.067	0.008	0.005	0.150	-0.103	-0.057
Female	-0.041	0.213	-0.095	-0.106	0.202	0.390	0.025	-0.216
Married	0.291	-0.027	0.014	-0.026	-0.151	0.055	-0.253	0.270
Any kid in HH	0.066	-0.088	0.137	-0.225	-0.060	0.038	-0.117	-0.005
Bachelor's degree or more	0.162	-0.071	-0.152	0.009	-0.055	-0.217	0.057	-0.108
Good self-reported health	0.124	0.104	0.190	0.134	-0.270	-0.044	-0.285	-0.231
Daily prescription	-0.107	0.090	-0.194	-0.136	0.265	0.148	0.031	0.082
Lives in a Medicaid expansion state	0.147	0.389	-0.146	0.244	0.113	-0.003	-0.030	0.144

Note: The table reports survey-weighted normalized differences between treated and control workers within each displacement cohort and SIPP panel. Values exceeding ±0.25 are shown in red.

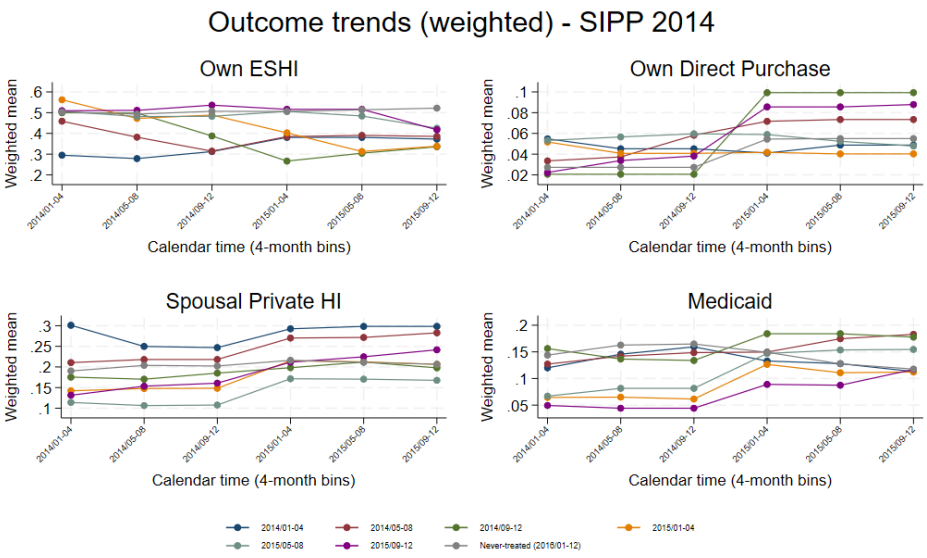
Table 2: Duration-specific effects of job displacement on health insurance coverage: pooled post-ACA sample

	Own ESHI	Own direct purchase	Spouse private	Medicaid	Any HI	Uninsured
$k = -1$	-0.004	0.000	0.004	0.006	0.011**	-0.011**
	(0.007)	(0.003)	(0.004)	(0.004)	(0.005)	(0.005)
$k = 0$	-0.006	-0.002	0.004	0.011**	0.016**	-0.016**
	(0.008)	(0.005)	(0.005)	(0.005)	(0.008)	(0.008)
$k = 1$	-0.117***	0.000	0.006	0.018**	-0.070***	0.070***
	(0.019)	(0.007)	(0.008)	(0.007)	(0.017)	(0.017)
$k = 3$	-0.101***	0.013	-0.000	0.026***	-0.034	0.034
	(0.024)	(0.009)	(0.011)	(0.010)	(0.021)	(0.021)
$k = 6$	-0.065**	0.006	0.008	0.034**	0.013	-0.013
	(0.029)	(0.012)	(0.015)	(0.013)	(0.025)	(0.025)
$k = 9$	-0.054	0.009	0.019	0.054***	0.067**	-0.067**
	(0.036)	(0.019)	(0.024)	(0.018)	(0.029)	(0.029)
$k = 12$	-0.032	0.014	0.008	0.059**	0.082**	-0.082**
	(0.051)	(0.024)	(0.039)	(0.024)	(0.033)	(0.033)

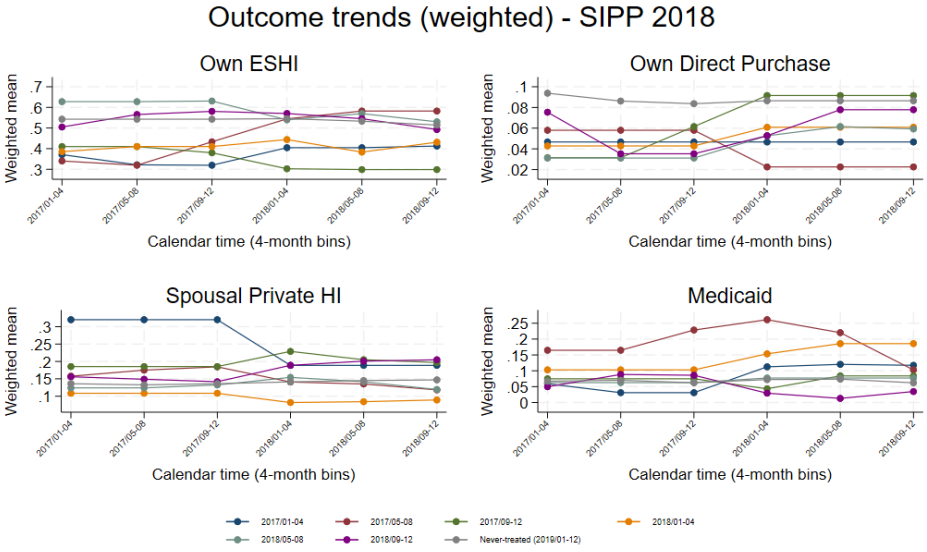
Note: This table reports covariate-adjusted event-study estimates of the effect of job displacement on health insurance coverage for the pooled SIPP 2014 and 2018 sample. k denotes months relative to displacement, and coefficients are expressed as shares. Estimates use individual survey weights normalized to mean one within each panel. Standard errors clustered at the individual level are reported in parentheses. The balanced sample contains 852 workers observed for 24 months, totaling 20,448 person-month observations. Estimation retains all observations within this period for workers displaced after the estimation window, and only observations at $k = -6, \ldots, 12$ for workers displaced within the window. This restriction leaves 14,689 person-month observations for every outcome. The four insurance-source indicators are not mutually exclusive. Any health insurance and uninsurance are complementary. Sample and estimation details are described in Sections 2 and 3. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Figure 1: Coverage rates over time by cohort

Panel A: SIPP 2014

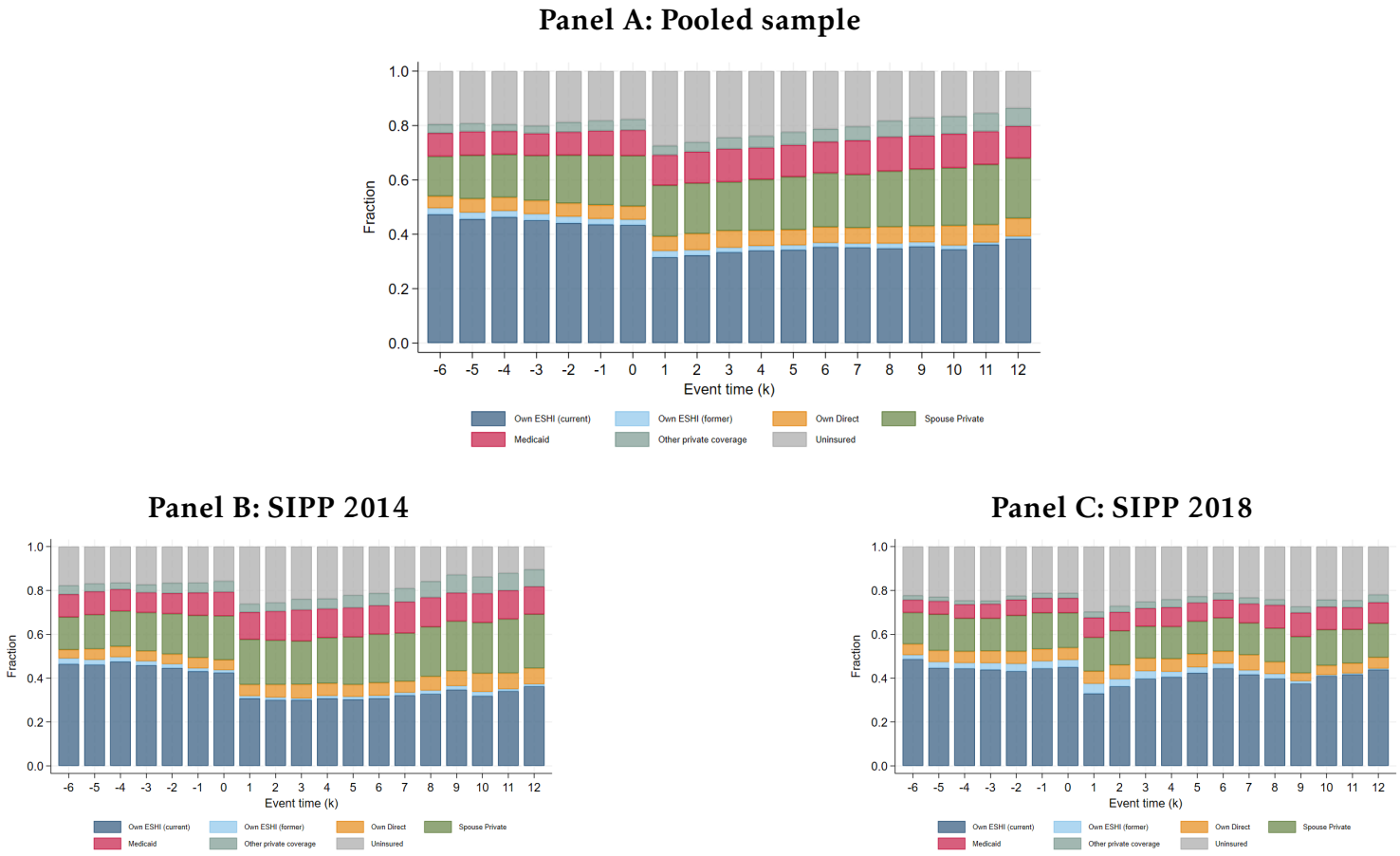


Panel B: SIPP 2018



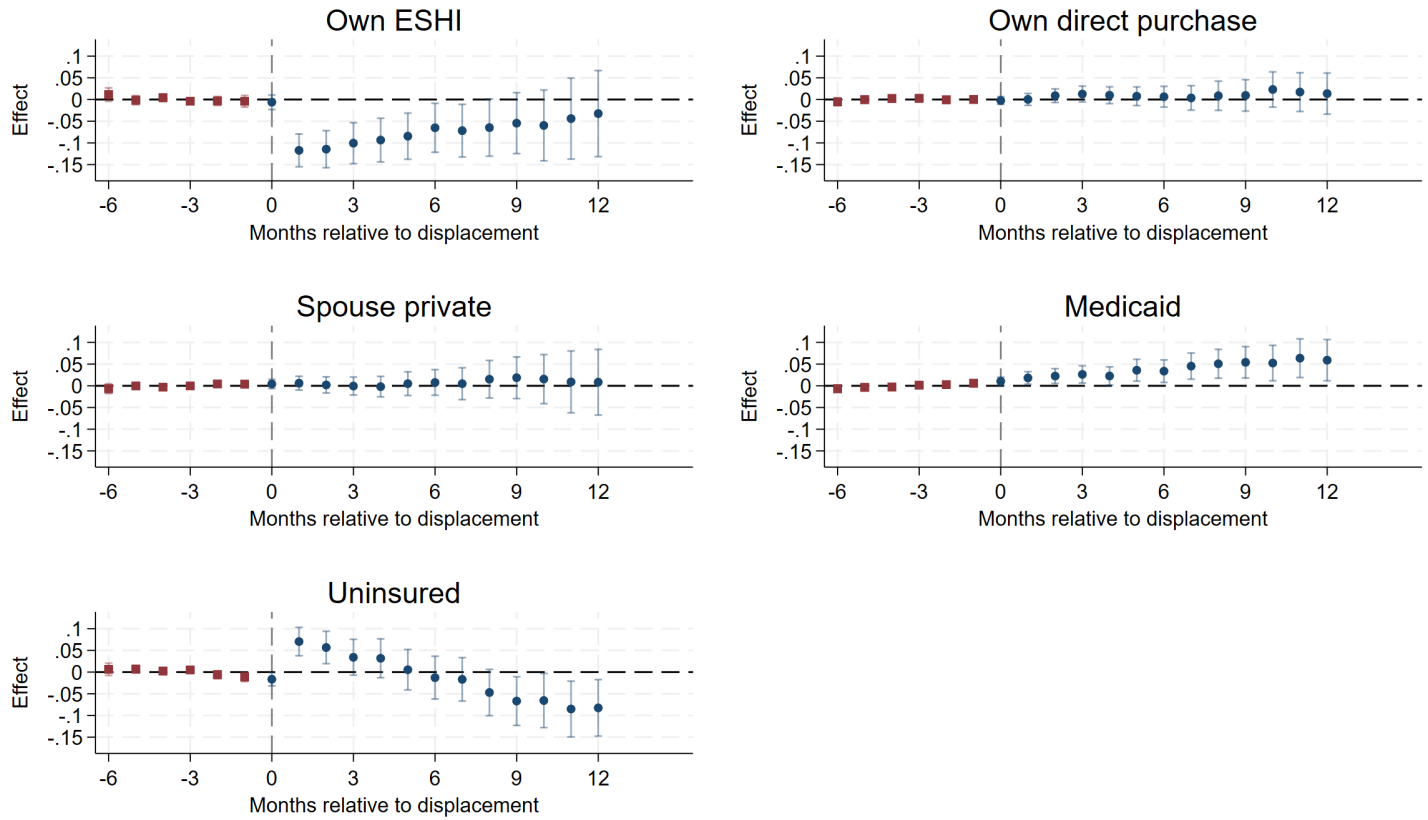
Note: The figure plots weighted mean insurance coverage by four-month calendar period and displacement cohort. Treated cohorts are defined by the interval in which displacement occurs. Never-treated workers are displaced after the estimation window and remain untreated during that window. Means use individual survey weights normalized within panel. The four coverage outcomes are not mutually exclusive. Workers never displaced during the panel are excluded.

Figure 2: Insurance states by event time: pooled and panel-specific post-ACA samples



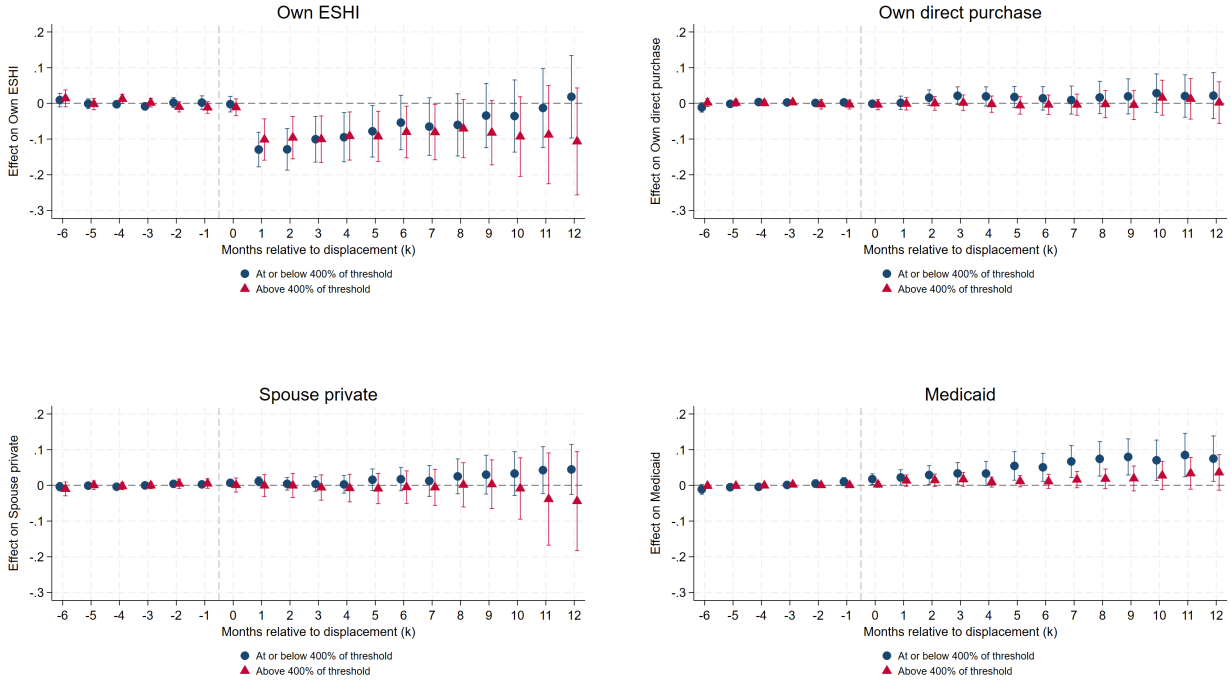
Note: The figure shows survey-weighted insurance-state shares by month relative to displacement among workers displaced within the estimation window. Panel A pools SIPP 2014 and 2018. Panels B and C report the two survey panels separately. Insurance states are mutually exclusive and are assigned in the following order: own ESHI from a current employer, own ESHI from a former employer, own direct-purchase coverage, spouse private coverage, Medicaid, other private coverage, and uninsured. Other private coverage comprises private insurance when none of the preceding sources applies. Uninsured means no private insurance or Medicaid. Shares use individual survey weights. COBRA coverage is expected to be reported as former-employer coverage, although SIPP does not identify COBRA directly. Regression outcomes are non-exclusive, and the workers observed at each event time may differ.

Figure 3: Duration-specific effects of job displacement on health insurance coverage: pooled post-ACA sample



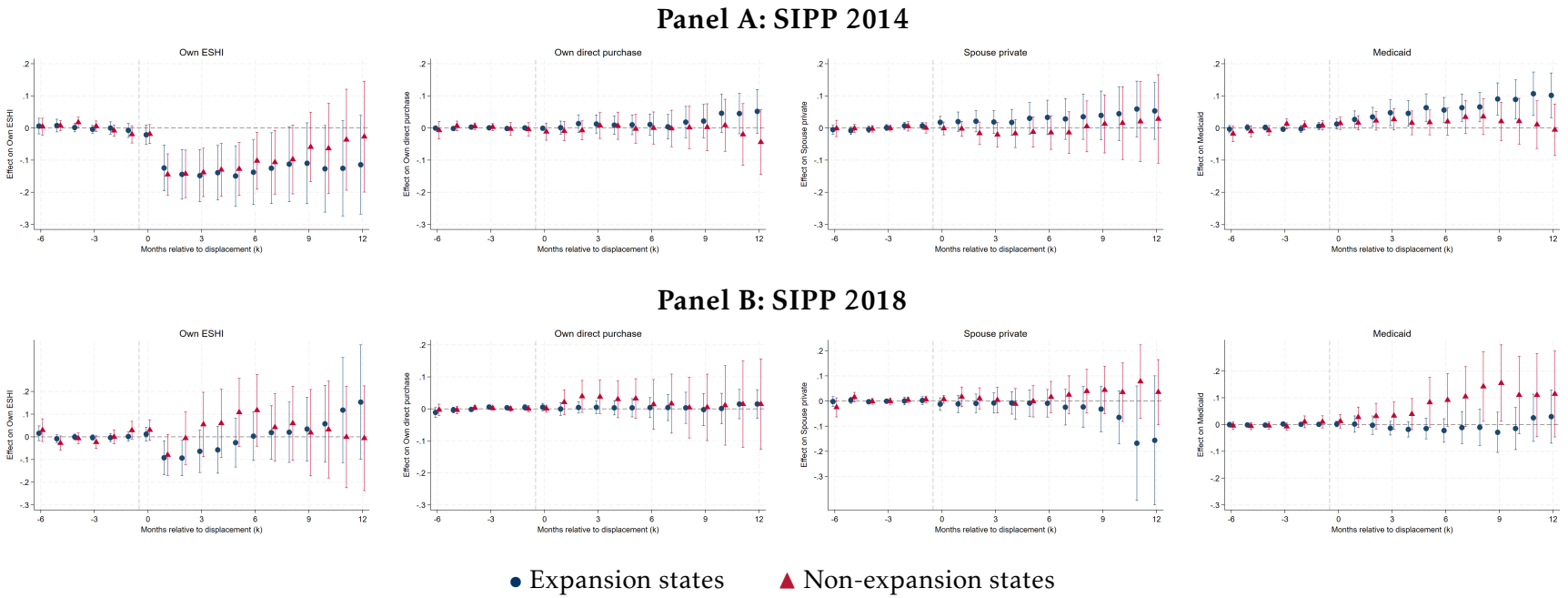
Note: This figure reports covariate-adjusted event-study estimates of the effect of job displacement on health insurance coverage. Estimates use individual survey weights normalized to mean one within each panel. Pointwise 95% confidence intervals are based on standard errors clustered at the individual level. The four insurance-source indicators are not mutually exclusive. Sample and estimation details are described in Sections 2 and 3.

Figure 4: Duration-specific effects of job displacement on health insurance coverage by pre-displacement household income: pooled post-ACA sample



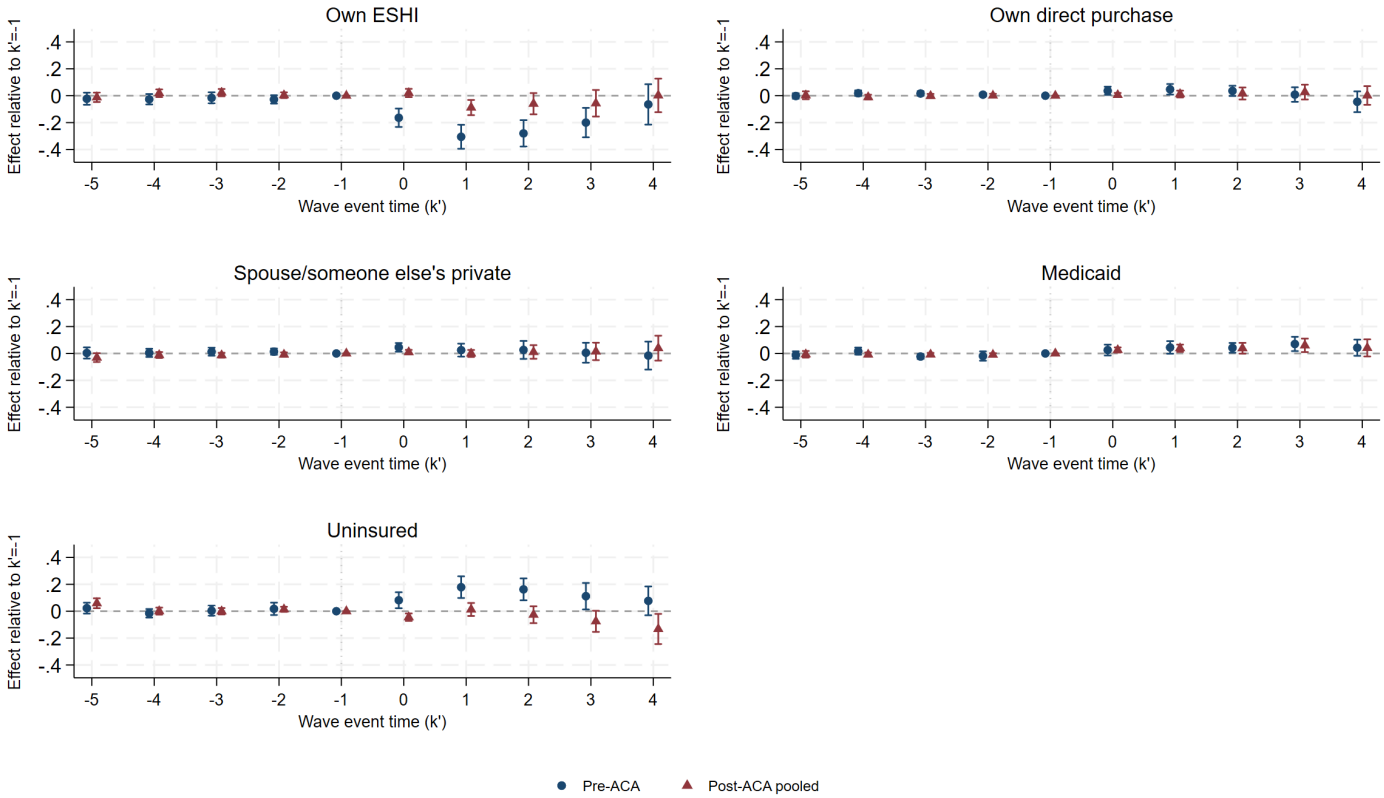
Note: This figure reports covariate-adjusted event-study estimates by pre-displacement household income for the pooled SIPP 2014 and 2018 sample. Workers are grouped by whether total monthly household income at $g-1$ exceeds 400% of the Census poverty threshold for their household. Estimates use individual survey weights normalized to mean one within each panel. Pointwise 95% confidence intervals are based on standard errors clustered at the individual level. The four insurance-source outcomes are not mutually exclusive.

Figure 5: Duration-specific effects of job displacement on health insurance coverage by Medicaid expansion status: post-ACA survey panels



Note: Panels A and B report covariate-adjusted event-study estimates of the effect of job displacement on health insurance coverage by Medicaid expansion status in SIPP 2014 and 2018, respectively. Expansion status is measured in the month before displacement ($g-1$). Estimates use individual survey weights normalized to mean one within each panel. Pointwise 95% confidence intervals are based on standard errors clustered at the individual level. The four insurance-source outcomes are not mutually exclusive. Sample and estimation details are described in Sections 2 and 3.

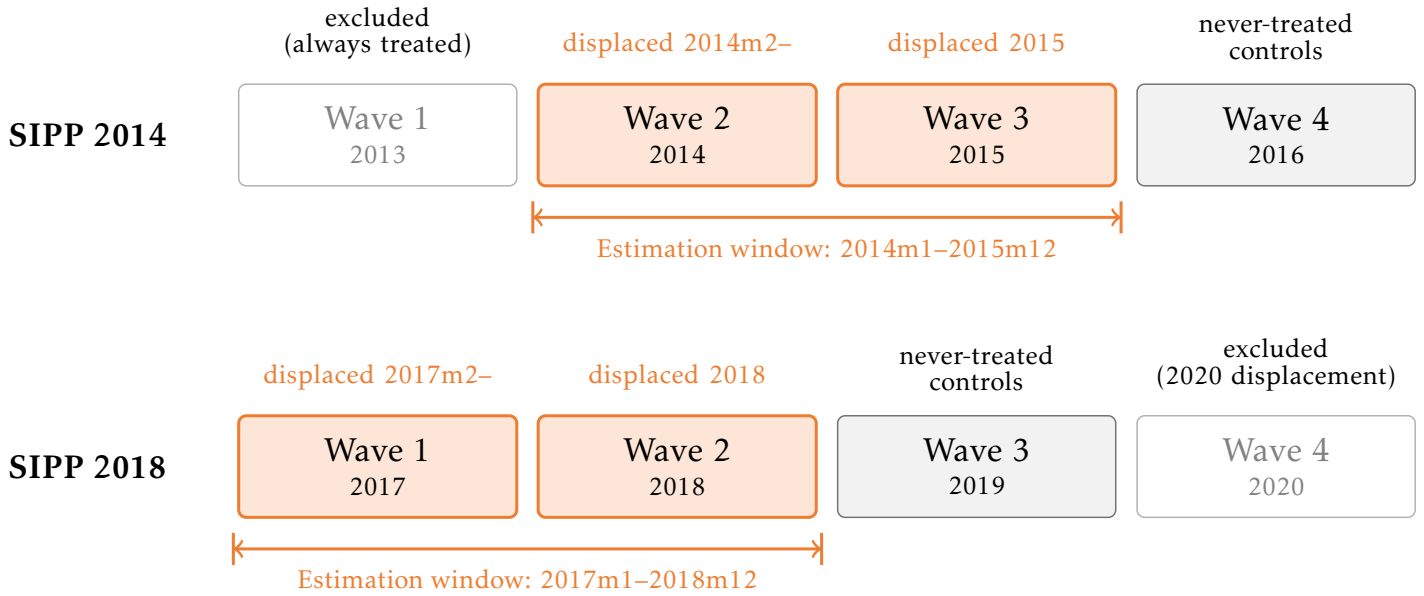
Figure 6: Duration-specific effects of job displacement on health insurance coverage: pre- and post-ACA samples



Note: This figure reports covariate-adjusted wave-level event-study estimates for the pre- and post-ACA samples. The pre-ACA series uses SIPP 2008 waves, and the post-ACA series uses four-month bins from pooled SIPP 2014 and 2018. Estimates use the longitudinal survey weights specified for each panel, with post-ACA weights normalized to mean one within panel. Pointwise 95% confidence intervals are based on standard errors clustered at the individual level. Estimates are recentered at $k' = -1$. Uninsured means no coverage in any observed month of the wave.

Appendix

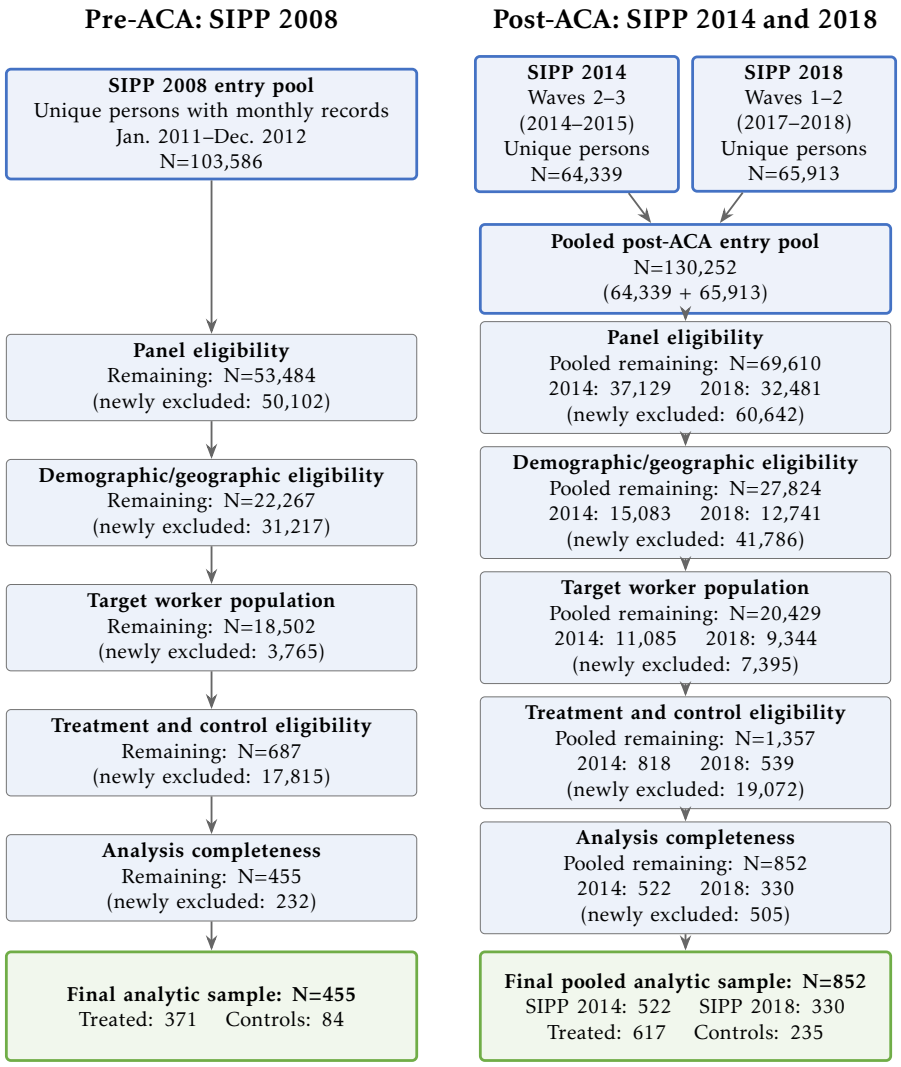
Figure A1: Sample construction and estimation windows



Workers never displaced during the panel are excluded from both samples.

Note: This figure illustrates the sample construction and estimation windows for the two post-ACA SIPP panels. Colors distinguish cohorts included as treated workers, cohorts used as controls, and cohorts excluded from the analysis. The SIPP 2014 panel spans waves 1–4 (2013–2016), with an estimation window from January 2014 through December 2015. Workers displaced in wave 1 are excluded to avoid always-treated cases. Workers first displaced in wave 4 contribute to the control group because they remain untreated throughout the estimation window. The SIPP 2018 panel spans waves 1–4 (2017–2020), with an estimation window from January 2017 through December 2018. Workers first displaced in wave 3 contribute to the control group because they remain untreated throughout the estimation window. Workers first displaced in wave 4 are excluded because that wave covers 2020 and overlaps the COVID-19 period. Workers never displaced during the panel are excluded from both samples.

Figure A2: Sample selection for the pre- and post-ACA analytic samples



Note: The flowchart reports cumulative sample selection for the pre-ACA SIPP 2008 panel and the pooled post-ACA SIPP 2014 and 2018 panels. All counts refer to unique workers. The counts shown precede the additional exclusions for the wave-level analysis in the pre/post-ACA comparison in Section 5. This analysis excludes workers displaced in their first observed wave in both samples. In the pre-ACA sample, workers with zero longitudinal survey weight also do not contribute to estimation. Panel eligibility requires observation throughout the panel-specific analysis window. The demographic and geographic eligibility stage limits the sample to workers ages 26–59 residing in the 50 states or the District of Columbia. The target-worker restrictions exclude retirees, the self-employed, workers with Medicare, military, or other public health insurance, and workers with active-duty or military employment. Treatment and control eligibility retains workers first displaced within the panel-specific treatment or control cohort windows. Workers first displaced during the estimation window form the treated group. Workers first displaced afterward form the control group and remain untreated throughout the estimation window. Workers already displaced when the window begins and workers never displaced during the panel are excluded. Analysis completeness requires nonmissing estimation variables. "Newly excluded" counts workers who fail at least one restriction at a given stage among those retained after the preceding stage.

Table A1: Separation reasons among treated workers

Separation reason	Individuals	Share (%)
Plant or company closed down or moved	133	21.6
Slack work or business conditions	200	32.4
Position or shift abolished	99	16.0
Other involuntary reason	185	30.0
Imputed (reason missing)	0	0.0
Total	617	100.0

Notes: The denominator is workers displaced within the analysis window: 617 in the pooled post-ACA sample. Imputed status is reported separately from observed separation reasons.

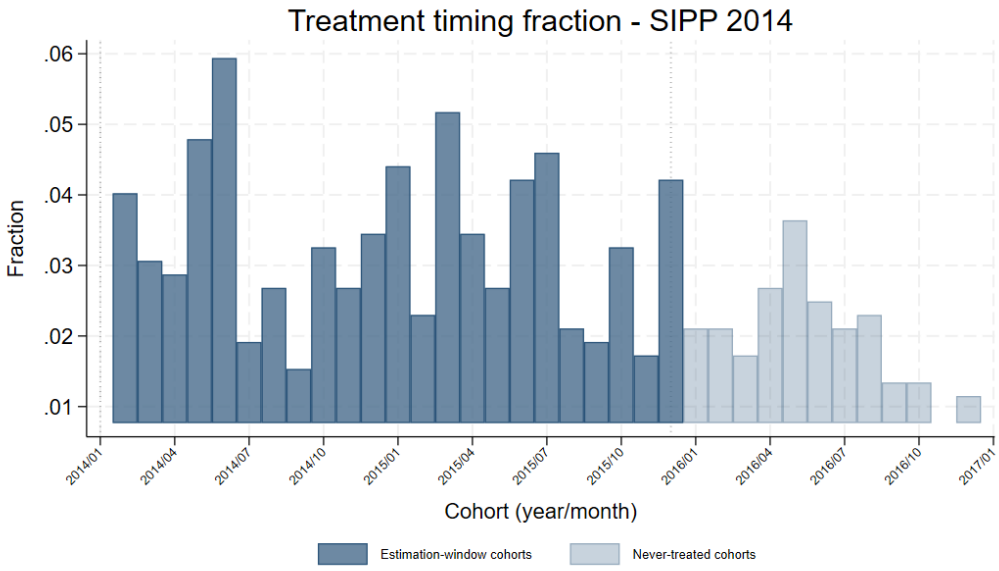
Table A2: Classification of health insurance coverage

Coverage category	Policyholder		
	Own	Spouse	Other person
Uninsured (no coverage of any kind)			
Any insurance			
Private (by source)			
Current employer	**Own ESHI***	**Spouse private***	†
Former employer			
Directly purchased	**Own DPHI***		
Union / professional assoc.			
School			
Medicaid*	(no policyholder distinction)		
Excluded from the estimation sample if held at any point during the sample period			
Medicare			
Military (TRICARE/CHAMPUS/CHAMPVA/VA)			
Other public coverage			

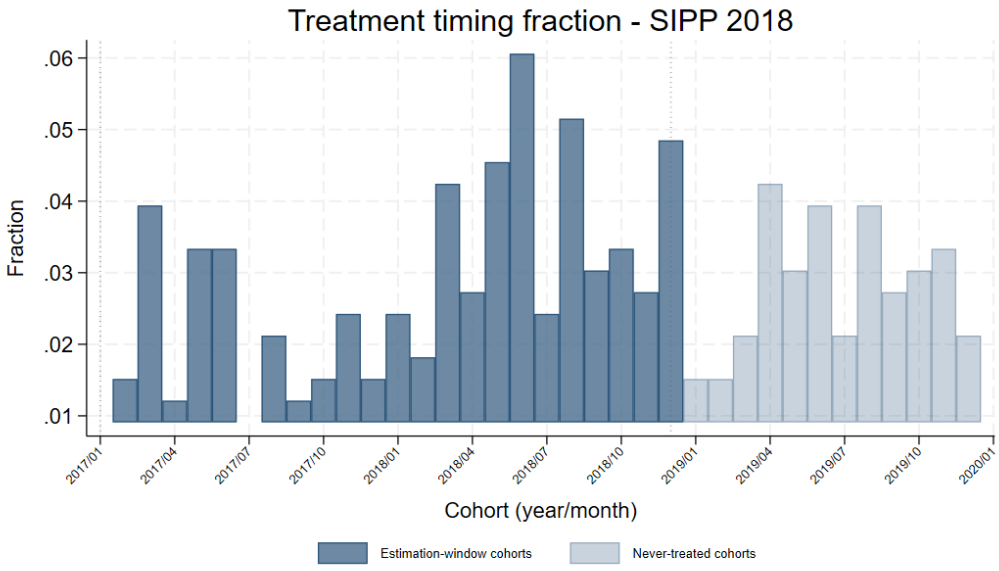
Notes: The four starred source indicators are not mutually exclusive. Spouse private covers private sources held in the spouse's name. Shaded categories are excluded by sample restrictions. Any coverage is the union of private coverage and Medicaid in the analysis sample. Uninsurance is its complement.

Figure A3: Distribution of displacement timing

Panel A: SIPP 2014



Panel B: SIPP 2018



Note: This figure shows the unweighted distribution of workers across monthly displacement cohorts, separately for SIPP 2014 and 2018. Dark bars identify cohorts displaced within the estimation window. Light bars identify workers displaced after the estimation window who remain untreated during that window and contribute to the control group. Fractions sum to one within each panel across the displayed cohorts. Workers never displaced during the panel are excluded.

Table A3: Changes in covariates by cohort

	SIPP 2014				SIPP 2018			
Covariate	2014/02–06	2014/07–12	2015/01–06	2015/07–12	2017/02–06	2017/07–12	2018/01–06	2018/07–12
Covariates likely affected by treatment								
Δ Employed	-0.791	-0.883	-0.826	-0.895	-0.909	-0.804	-0.779	-0.783
Δ Earnings (if employed)	0.037	-0.144	-0.146	-0.124	0.156	-0.119	0.097	-0.537
Δ Hours (if employed)	-0.205	-0.529	-0.280	-0.612	0.077	-0.256	-0.482	-0.522
Δ Full-time (if employed)	-0.010	-0.481	-0.281	-0.593	-0.014	-0.544	-0.394	-0.458
Δ Firm size >50 (if employed)	-0.417	0.384	-0.041	0.036	-0.510	-0.148	0.516	0.160
Δ Service sector (if employed)	-0.081	0.367	0.134	0.056	-0.326	-0.021	0.180	0.013
Δ Household income-to-poverty ratio	-0.351	-0.485	-0.384	-0.554	0.144	-0.242	-0.384	-0.576
Δ Spouse employed (if married)	-0.076	-0.266	-0.094	-0.214	-0.026	0.395	-0.310	-0.178
Δ Spouse full-time (if married & spouse employed)	-0.154	-0.070	-0.264	0.182	0.119	-0.008	-0.185	-0.285
Δ Lives in a Medicaid expansion state	-0.007	-0.082	-0.156	-0.110	0.084	-0.069	-0.216	.
Other covariates								
Δ Age	-0.060	-0.078	-0.010	-0.137	-0.375	0.133	0.175	-0.052
Δ Female	.	0.029	0.034	.	.	-0.028	-0.141	.
Δ Married	0.004	-0.007	0.043	-0.110	.	-0.078	-0.202	0.068
Δ Any kid in HH	0.035	0.069	-0.125	0.091	-0.049	-0.054	0.154	-0.007
Δ Bachelor's degree or more	.	0.057	0.067	.	.	.	.	.
Δ Good self-reported health	.	-0.101	-0.115	.	.	-0.013	0.128	.
Δ Daily prescription	.	0.169	-0.088	.	.	-0.017	0.042	.

Note: The table reports survey-weighted normalized differences in changes from one month before to one month after displacement. Values exceeding ±0.25 are shown in red. Changes in covariates likely affected by displacement are not used to select adjustment variables.

Table A4: Transitions from own ESHI before displacement

	$k = 0$	$k = 3$	$k = 6$	$k = 12$
Own ESHI (current)	0.9465	0.6603	0.6898	0.6218
Own ESHI (former)	0.0425	0.0379	0.0359	0.0218
Own Direct	0.0000	0.0354	0.0371	0.0562
Spouse Private	0.0062	0.0376	0.0638	0.1384
Medicaid	0.0026	0.0472	0.0247	0.0078
Other private coverage	0.0000	0.0075	0.0184	0.0812
Uninsured	0.0022	0.1741	0.1303	0.0729

Note: The table reports survey-weighted insurance-state shares at selected months after displacement for workers in the pooled SIPP 2014 and 2018 sample who had own ESHI from a current or former employer at $k = -1$ and are observed at the indicated event time. Shares sum to one within each column. Insurance states are mutually exclusive and are assigned in the following order: own ESHI from a current employer, own ESHI from a former employer, own direct-purchase coverage, spouse private coverage, Medicaid, other private coverage, and uninsured. Other private coverage comprises private insurance when none of the preceding sources applies. Uninsured means no private insurance or Medicaid. COBRA coverage is expected to be reported as former-employer coverage, although SIPP does not identify COBRA directly. Weights are normalized to mean one within each survey panel. The workers represented may differ across columns because observation at the indicated event time is required.

Table A5: Event-time support: main specification

k	Panel	N treated	N control	N ctrl not-yet	N ctrl out-window	Treated weight	$\omega_p(k)$
-6	2014	290	4097	1865	2232	275.667	0.605
-6	2018	175	3004	1006	1998	180.142	0.395
-5	2014	321	4290	1934	2356	306.749	0.617
-5	2018	186	3178	1069	2109	190.250	0.383
-4	2014	346	4472	1992	2480	332.233	0.623
-4	2018	197	3335	1115	2220	201.450	0.377
-3	2014	361	4644	2040	2604	347.069	0.629
-3	2018	201	3482	1151	2331	204.406	0.371
-2	2014	377	4799	2071	2728	366.786	0.629
-2	2018	214	3618	1176	2442	216.056	0.371
-1	2014	398	4945	2093	2852	396.721	0.642
-1	2018	219	3745	1192	2553	221.421	0.358
0	2014	398	4827	1975	2852	396.721	0.642
0	2018	219	3698	1145	2553	221.421	0.358
1	2014	376	4592	1864	2728	377.025	0.645
1	2018	203	3538	1096	2442	207.670	0.355
2	2014	367	4365	1761	2604	365.558	0.647
2	2018	194	3387	1056	2331	199.142	0.353
3	2014	350	4136	1656	2480	352.163	0.652
3	2018	183	3235	1015	2220	187.874	0.348
4	2014	340	3918	1562	2356	341.584	0.661
4	2018	173	3086	977	2109	175.460	0.339
5	2014	329	3713	1481	2232	327.906	0.671
5	2018	156	2943	945	1998	160.551	0.329
6	2014	305	3495	1387	2108	307.845	0.668
6	2018	148	2795	908	1887	152.922	0.332
7	2014	283	3279	1295	1984	286.309	0.687
7	2018	128	2648	872	1776	130.220	0.313
8	2014	269	3044	1184	1860	274.149	0.708
8	2018	113	2491	826	1665	113.007	0.292
9	2014	251	2808	1072	1736	258.902	0.713
9	2018	104	2330	776	1554	104.204	0.287
10	2014	224	2572	960	1612	231.116	0.724
10	2018	90	2162	719	1443	87.903	0.276
11	2014	212	2332	844	1488	219.894	0.728
11	2018	84	1979	647	1332	82.321	0.272
12	2014	189	2091	727	1364	198.979	0.729
12	2018	76	1796	575	1221	74.155	0.271

Notes: This table reports event-time support for the main pooled post-ACA specification. k denotes months relative to displacement. Treated counts are persons observed at k. Control counts are untreated person-months in the corresponding calendar months and are divided into not-yet-treated and out-of-window observations. Treated weight is the sum of normalized survey weights among treated persons. $\omega_p(k)$ is the panel's share of total treated weight at each k, which is used to aggregate the panel-specific estimates.

Table A6: Joint tests of pre-displacement coefficients

Panel A: Gardner specification

	Own ESHI	Own Direct Purchase	Spouse private	Medicaid
Pretrend joint Wald chi2	6.198	3.034	3.736	2.756
p-value	0.287	0.695	0.588	0.738

Panel B: Borusyak et al. specification

	Own ESHI	Own Direct Purchase	Spouse private	Medicaid
Pretrend joint Wald chi2	9.063	2.266	9.096	5.587
p-value	0.170	0.894	0.168	0.471

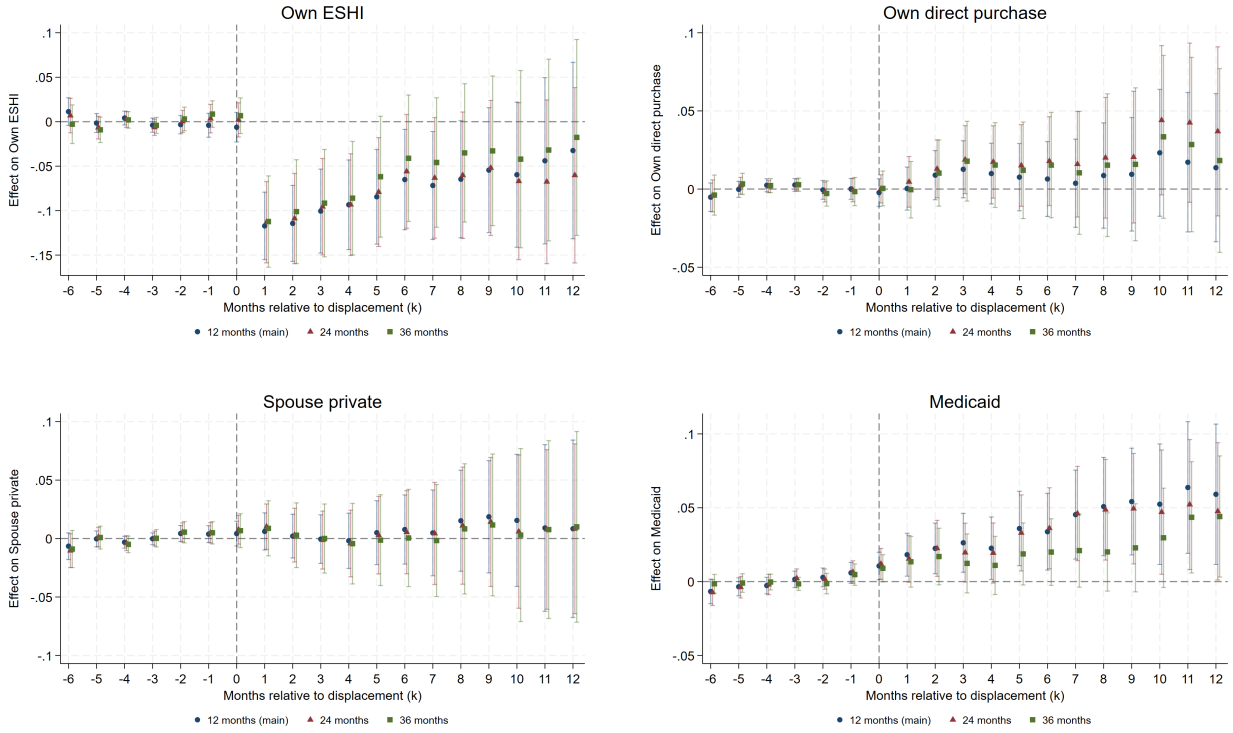
Note: The table reports joint Wald tests of the pre-displacement coefficients for the four insurance-source outcomes in the pooled SIPP 2014 and 2018 sample. Panel A uses the covariate-adjusted Gardner specification and tests $k = -6, \ldots, -2$, with $k = -1$ as the reference period. Its first stage uses control-group observations and observations at $k = -1$ for treated workers. Panel B uses the covariate-adjusted Borusyak et al. specification, estimates the model on untreated observations, and tests $k = -6, \ldots, -1$. Both models include individual and calendar-month fixed effects, panel-specific baseline covariates interacted with calendar-month indicators, and age as a time-varying covariate for SIPP 2018. Estimates use survey weights normalized within panel. Standard errors are clustered at the individual level. Reported p-values use the chi-squared reference distribution.

Table A7: Sensitivity of selected event-study estimates to violations of parallel trends

Outcome or sample	k	Estimate	Conventional 95% CI	Robust 95% CI, $\bar{M} = 1$	Breakdown $\bar{M}$
Panel A: Pooled post-ACA sample					
Own ESHI	3	-9.65	[-13.99, -5.30]	[-20.15, 0.85]	0.88
Own ESHI	12	-2.84	[-12.60, 6.93]	[-36.93, 28.90]	0
Medicaid	12	5.32	[0.69, 9.95]	[-11.21, 21.85]	0.12
Panel B: Medicaid, expansion states					
SIPP 2014	3	4.16	[0.41, 7.90]	[-4.67, 12.99]	0.12
	6	5.00	[1.29, 8.71]	[-9.60, 19.61]	0.24
	9	8.44	[3.57, 13.31]	[-12.13, 29.01]	0.38
	12	9.57	[2.77, 16.36]	[-17.25, 38.00]	0.32
SIPP 2018	3	-1.43	[-3.68, 0.81]	[-6.65, 4.28]	0
	6	-2.30	[-6.51, 1.91]	[-12.97, 8.37]	0
	9	-2.97	[-10.41, 4.46]	[-18.35, 10.79]	0
	12	2.88	[-7.03, 12.79]	[-15.43, 21.19]	0
Panel C: Pre-ACA estimates and pre-minus-post differences					
Pre-ACA own ESHI	0	-16.41	[-23.26, -9.56]	[-32.64, -0.18]	1.08
Pre-ACA own ESHI	1	-30.57	[-39.49, -21.64]	[-61.13, -0.01]	1.01
Pre–post own ESHI	1	-21.75	[-32.31, -11.19]	[-42.67, -2.22]	1.18
Pre–post uninsured	1	16.59	[7.17, 26.00]	[-0.88, 35.66]	0.93

Note: The table reports selected estimates and pointwise 95% confidence intervals in percentage points. Panel A uses the pooled post-ACA sample. Panel B reports Medicaid estimates for expansion states in SIPP 2014 and 2018. Expansion status is measured in the month before displacement. Panel C reports pre-ACA own ESHI estimates and differences between pre- and post-ACA estimates, calculated as pre minus post. Monthly estimates use $k = -1$ as the reference period. Wave-level estimates and pre-minus-post differences use $k' = -1$. Conventional intervals are Wald intervals based on standard errors clustered at the individual level. Robust intervals use the relative-magnitude restriction Δ^{RM}, method C-LF, $\bar{M} = 1$, and $\alpha = 0.05$. Breakdown values are the smallest $\bar{M}$ at which the robust interval includes zero. For pre-minus-post differences, the restriction applies to the difference in biases across periods.

Figure A4: Duration-specific effects of job displacement on health insurance coverage under alternative minimum tenure requirements: pooled post-ACA sample



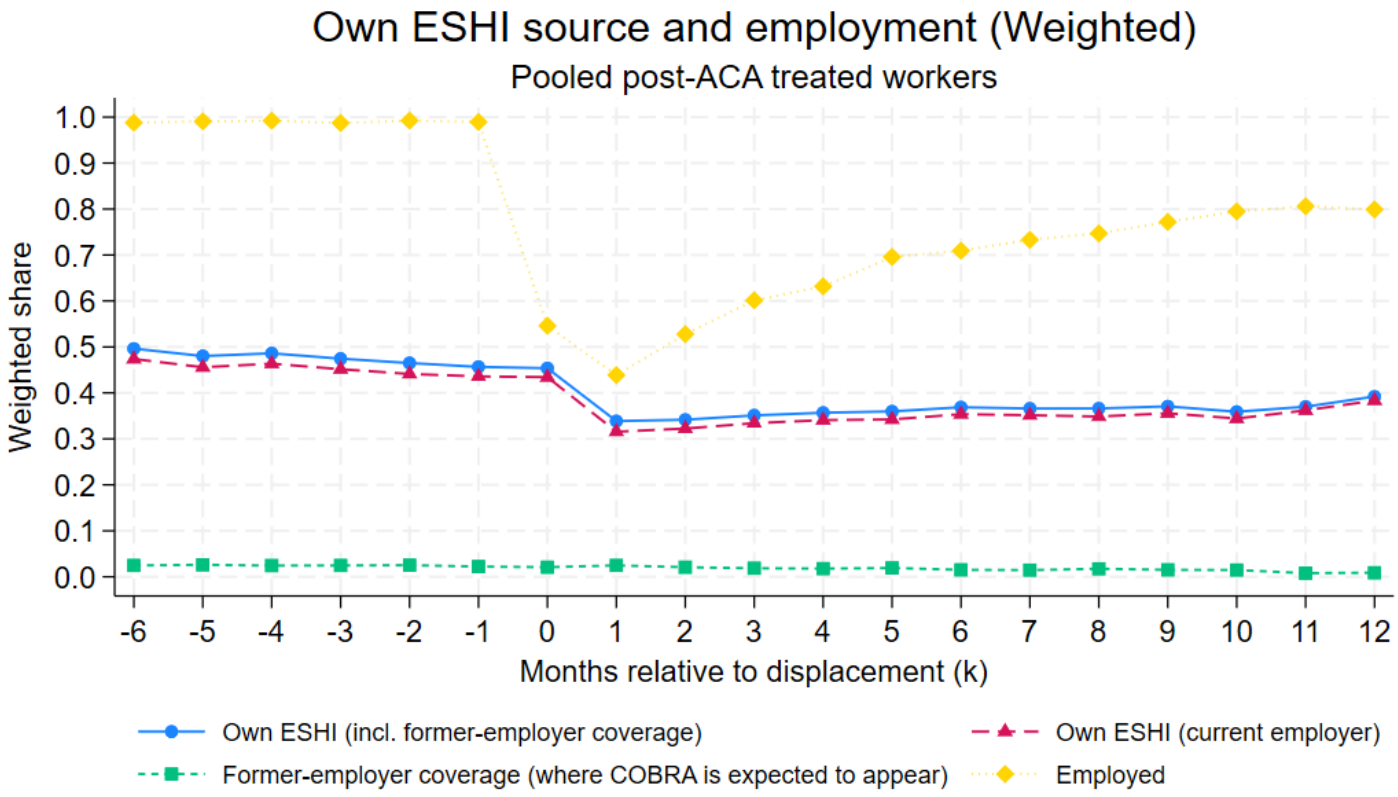
Note: This figure reports covariate-adjusted event-study estimates under 12-, 24-, and 36-month minimum tenure requirements. Estimates use individual survey weights normalized to mean one within each panel. Pointwise 95% confidence intervals are based on standard errors clustered at the individual level. The four insurance-source indicators are not mutually exclusive. The 36-month specification is a selected-sample robustness check because jobs with imputed start dates rarely meet this threshold.

Table A8: Effects under alternative minimum tenure requirements at selected horizons

Outcome	k	12 months	24 months	36 months
Own ESHI	1	-0.117	-0.113	-0.112
		(0.019)	(0.023)	(0.026)
	3	-0.101	-0.096	-0.092
		(0.024)	(0.028)	(0.031)
	6	-0.065	-0.056	-0.041
		(0.029)	(0.033)	(0.036)
	9	-0.054	-0.052	-0.033
		(0.036)	(0.039)	(0.043)
	12	-0.032	-0.060	-0.018
		(0.051)	(0.050)	(0.056)
Own direct purchase	1	0.000	0.005	-0.000
		(0.007)	(0.008)	(0.009)
	3	0.013	0.019	0.018
		(0.009)	(0.011)	(0.013)
	6	0.006	0.018	0.015
		(0.012)	(0.014)	(0.017)
	9	0.009	0.020	0.016
		(0.019)	(0.021)	(0.025)
	12	0.014	0.037	0.018
		(0.024)	(0.028)	(0.030)
Spouse private	1	0.006	0.010	0.009
		(0.008)	(0.010)	(0.012)
	3	-0.000	-0.001	0.000
		(0.011)	(0.013)	(0.015)
	6	0.008	0.005	0.001
		(0.015)	(0.018)	(0.021)
	9	0.019	0.014	0.012
		(0.024)	(0.028)	(0.031)
	12	0.008	0.008	0.010
		(0.039)	(0.037)	(0.042)
Medicaid	1	0.018	0.015	0.013
		(0.007)	(0.008)	(0.009)
	3	0.026	0.020	0.012
		(0.010)	(0.010)	(0.010)
	6	0.034	0.036	0.020
		(0.013)	(0.014)	(0.011)
	9	0.054	0.049	0.023
		(0.018)	(0.019)	(0.015)
	12	0.059	0.048	0.044
		(0.024)	(0.024)	(0.021)

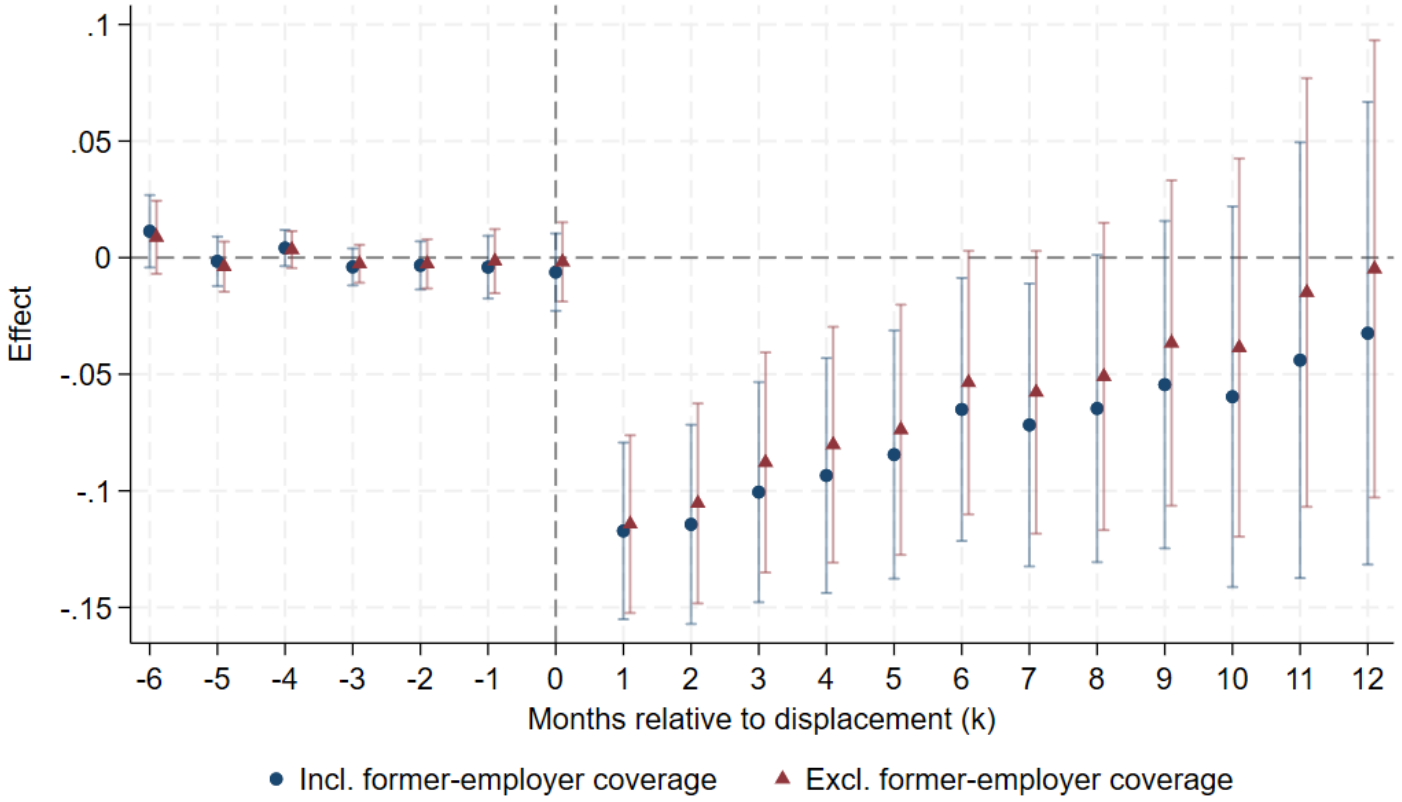
Note: The table reports covariate-adjusted estimates at selected post-displacement horizons. Each cell gives the coefficient with its individual-cluster standard error in parentheses. Coefficients are expressed as shares. The 24- and 36-month requirements retain 446 and 356 of the 617 treated workers in the 12-month sample. Full event-time paths are shown in Figure A4.

Figure A5: Current- and former-employer coverage



Note: This figure plots weighted mean coverage and employment by month relative to displacement for treated workers in the pooled post-ACA sample. Own ESHI includes coverage from both current and former employers. Current- and former-employer coverage can coexist in the same person-month. Former-employer coverage is where COBRA coverage is expected to be reported, although SIPP does not identify COBRA directly. These series are descriptive means rather than event-study estimates. Sample composition varies across event time.

Figure A6: Duration-specific effects of job displacement on own ESHI with and without former-employer coverage: pooled post-ACA sample



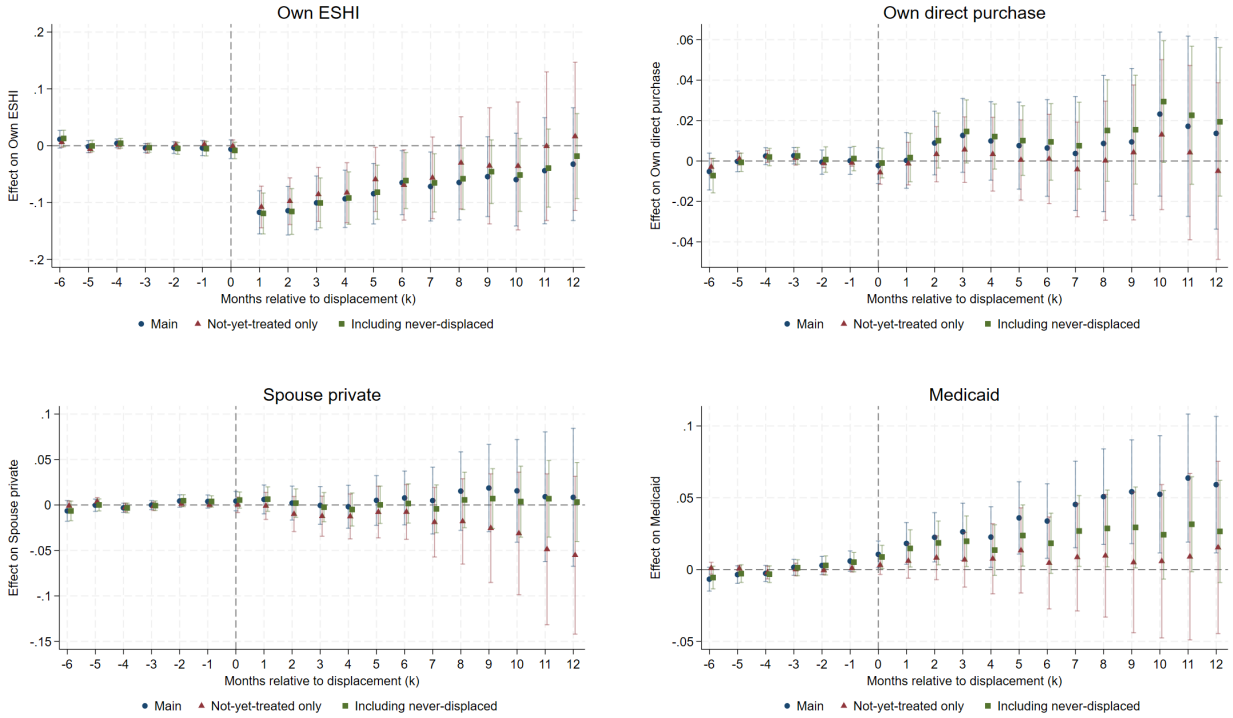
Note: This figure compares covariate-adjusted event-study estimates of the effect of job displacement on own ESHI under two coverage definitions for the pooled SIPP 2014 and 2018 sample. The main definition includes coverage held in the worker's own name from a current or former employer. The alternative definition includes only coverage from a current employer. Both series use the same estimation sample and covariate adjustment. Estimates use individual survey weights normalized to mean one within each panel. Pointwise 95% confidence intervals are based on standard errors clustered at the individual level. SIPP does not identify COBRA coverage directly, although it is expected to be reported as former-employer coverage.

Table A9: Event-study estimates for own ESHI with and without former-employer coverage: pooled post-ACA sample

	Incl. former-employer coverage	Excl. former-employer coverage
k=-6	0.011	0.009
	(0.008)	(0.008)
k=-5	-0.002	-0.004
	(0.005)	(0.005)
k=-4	0.004	0.003
	(0.004)	(0.004)
k=-3	-0.004	-0.003
	(0.004)	(0.004)
k=-2	-0.003	-0.003
	(0.005)	(0.005)
k=-1	-0.004	-0.002
	(0.007)	(0.007)
k=0	-0.006	-0.002
	(0.008)	(0.009)
k=1	-0.117***	-0.114***
	(0.019)	(0.019)
k=2	-0.114***	-0.105***
	(0.022)	(0.022)
k=3	-0.101***	-0.088***
	(0.024)	(0.024)
k=4	-0.093***	-0.080***
	(0.026)	(0.026)
k=5	-0.084***	-0.074***
	(0.027)	(0.027)
k=6	-0.065**	-0.054*
	(0.029)	(0.029)
k=7	-0.072**	-0.058*
	(0.031)	(0.031)
k=8	-0.065*	-0.051
	(0.034)	(0.034)
k=9	-0.054	-0.037
	(0.036)	(0.036)
k=10	-0.060	-0.039
	(0.042)	(0.041)
k=11	-0.044	-0.015
	(0.048)	(0.047)
k=12	-0.032	-0.005
	(0.051)	(0.050)
Observations	14689	14689

Note: This table reports the estimates underlying Figure A6. The main definition of own ESHI includes coverage held in the worker's own name from a current or former employer. The alternative definition includes only coverage from a current employer. Both specifications use the same estimation sample and covariate adjustment. k denotes months relative to displacement. Each cell gives the coefficient with its individual-cluster standard error in parentheses. Coefficients are expressed as shares. Estimates use individual survey weights normalized to mean one within each panel. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Figure A7: Duration-specific effects of job displacement on health insurance coverage under alternative comparison groups: pooled post-ACA sample



Note: This figure reports covariate-adjusted event-study estimates of the effect of job displacement on health insurance coverage under three comparison groups for the pooled SIPP 2014 and 2018 sample. The main specification uses pre-displacement observations from workers who have not yet been displaced and workers displaced after the estimation window as controls. The second specification excludes workers displaced after the estimation window and uses only pre-displacement observations from workers displaced within the window as controls. The final calendar month of each panel is excluded because no untreated observations remain. The third specification adds never-displaced workers to the main sample and restores workers displaced in 2020 in SIPP 2018. Covariates are selected separately within each sample. Estimates use individual survey weights normalized to mean one within each panel. Pointwise 95% confidence intervals are based on standard errors clustered at the individual level. The four insurance-source outcomes are not mutually exclusive.

Table A10: Effects of job displacement on health insurance coverage under alternative comparison groups at selected horizons: pooled post-ACA sample

Outcome	k	Main	Not-yet-treated only	Including never-displaced
Own ESHI	1	-0.117	-0.108	-0.119
		(0.019)	(0.019)	(0.018)
	3	-0.101	-0.086	-0.101
		(0.024)	(0.024)	(0.022)
	6	-0.065	-0.069	-0.061
		(0.029)	(0.032)	(0.025)
	9	-0.054	-0.035	-0.046
		(0.036)	(0.052)	(0.029)
	12	-0.032	0.016	-0.018
		(0.051)	(0.067)	(0.038)
Own direct purchase	1	0.000	-0.001	0.002
		(0.007)	(0.005)	(0.006)
	3	0.013	0.006	0.015
		(0.009)	(0.008)	(0.008)
	6	0.006	0.001	0.009
		(0.012)	(0.011)	(0.010)
	9	0.009	0.004	0.016
		(0.019)	(0.017)	(0.014)
	12	0.014	-0.005	0.019
		(0.024)	(0.022)	(0.019)
Spouse private	1	0.006	-0.001	0.006
		(0.008)	(0.008)	(0.007)
	3	-0.000	-0.012	-0.002
		(0.011)	(0.011)	(0.008)
	6	0.008	-0.008	0.002
		(0.015)	(0.015)	(0.011)
	9	0.019	-0.025	0.007
		(0.024)	(0.030)	(0.017)
	12	0.008	-0.055	0.003
		(0.039)	(0.044)	(0.022)
Medicaid	1	0.018	0.006	0.015
		(0.007)	(0.006)	(0.007)
	3	0.026	0.007	0.020
		(0.010)	(0.010)	(0.009)
	6	0.034	0.005	0.018
		(0.013)	(0.016)	(0.011)
	9	0.054	0.005	0.029
		(0.018)	(0.025)	(0.014)
	12	0.059	0.015	0.027
		(0.024)	(0.031)	(0.018)

Note: This table reports covariate-adjusted event-study estimates at selected post-displacement horizons for the pooled SIPP 2014 and 2018 sample. The main specification uses pre-displacement observations from workers who have not yet been displaced and workers displaced after the estimation window as controls. The second specification excludes workers displaced after the estimation window and uses only pre-displacement observations from workers displaced within the window as controls. The final calendar month of each panel is excluded because no untreated observations remain. The third specification adds never-displaced workers to the main sample and restores workers displaced in 2020 in SIPP 2018. Covariates are selected separately within each sample. Each cell gives the coefficient with its individual-cluster standard error in parentheses. Coefficients are expressed as shares. Estimates use individual survey weights normalized to mean one within each panel. The four insurance-source outcomes are not mutually exclusive. Full event-time paths are shown in Figure A7.

Table A11: Event-time support under alternative comparison groups at selected horizons: post-ACA survey panels

		SIPP 2014		SIPP 2018	
Comparison group	k	Treated	Controls	Treated	Controls
Not-yet-treated only	1	367	1,864	194	1,096
	6	283	1,387	128	908
	12	171	727	71	575
Including never-displaced	1	376	221,754	203	186,424
	6	305	171,302	148	144,116
	12	189	110,672	76	93,239

Note: This table reports sample support at $k = 1, 6$, and 12 for two alternative comparison groups in SIPP 2014 and 2018. The first specification excludes workers displaced after the estimation window and uses only pre-displacement observations from workers displaced within the window as controls. The second adds never-displaced workers to the main sample and restores workers displaced in 2020 in SIPP 2018. Treated counts are workers observed at the indicated event time. Control counts are untreated person-month observations in the calendar months contributing to the corresponding coefficient. A worker may contribute control observations to more than one event-time coefficient, so the control sets overlap across k.

Table A12: Effects and event-time support by pre-displacement household income: pooled post-ACA sample

Panel A: Event-study estimates at selected horizons

Sample	k	Own ESHI	Own direct purchase	Spouse private	Medicaid
≤ 400% of threshold	1	-0.129***	0.001	0.011*	0.022**
		(0.025)	(0.010)	(0.006)	(0.011)
	3	-0.101***	0.021*	0.004	0.034**
		(0.033)	(0.013)	(0.010)	(0.016)
	9	-0.034	0.020	0.030	0.080***
		(0.046)	(0.025)	(0.028)	(0.026)
	12	0.018	0.022	0.045	0.075**
		(0.059)	(0.033)	(0.036)	(0.032)
> 400% of threshold	1	-0.101***	-0.001	-0.001	0.013
		(0.030)	(0.009)	(0.016)	(0.008)
	3	-0.101***	0.002	-0.006	0.017*
		(0.033)	(0.011)	(0.018)	(0.010)
	9	-0.082*	-0.004	0.003	0.019
		(0.046)	(0.021)	(0.035)	(0.018)
	12	-0.107	0.002	-0.044	0.036
		(0.076)	(0.030)	(0.071)	(0.025)

Panel B: Treated workers and control observations by event time

	Income ≤ 400% of threshold		Income > 400% of threshold	
k	Treated workers	Control observations	Treated workers	Control observations
-6	265	7,101	200	6,958
-5	294	7,468	213	7,320
-4	316	7,807	227	7,660
-3	325	8,126	237	7,969
-2	338	8,417	253	8,256
-1	352	8,690	265	8,522
0	352	8,525	265	8,342
1	336	8,130	243	7,947
2	323	7,752	238	7,558
3	308	7,371	225	7,181
4	295	7,004	218	6,812
5	281	6,656	204	6,470
6	268	6,290	185	6,108
7	245	5,927	166	5,753
8	227	5,535	155	5,378
9	212	5,138	143	4,991
10	189	4,734	125	4,598
11	178	4,311	118	4,184
12	161	3,887	104	3,776

Note: Workers are grouped by whether total monthly household income at $g-1$ exceeds 400% of the Census poverty threshold for their household. Panel A reports covariate-adjusted coefficients with individual-cluster standard errors in parentheses. Coefficients are expressed as shares. Panel B reports counts, which are identical across the four insurance-source outcomes. Control observations are untreated person-months in the calendar months contributing to each subgroup's event-time estimate, not unique workers. The first-stage comparison pool is shared, but the contributing calendar months can differ between subgroups. The four insurance-source outcomes are not mutually exclusive. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Full event-time paths are shown in Figure 4.

Table A13: Effects and Medicaid support by expansion status and survey panel at selected horizons

Panel A: Event-study estimates

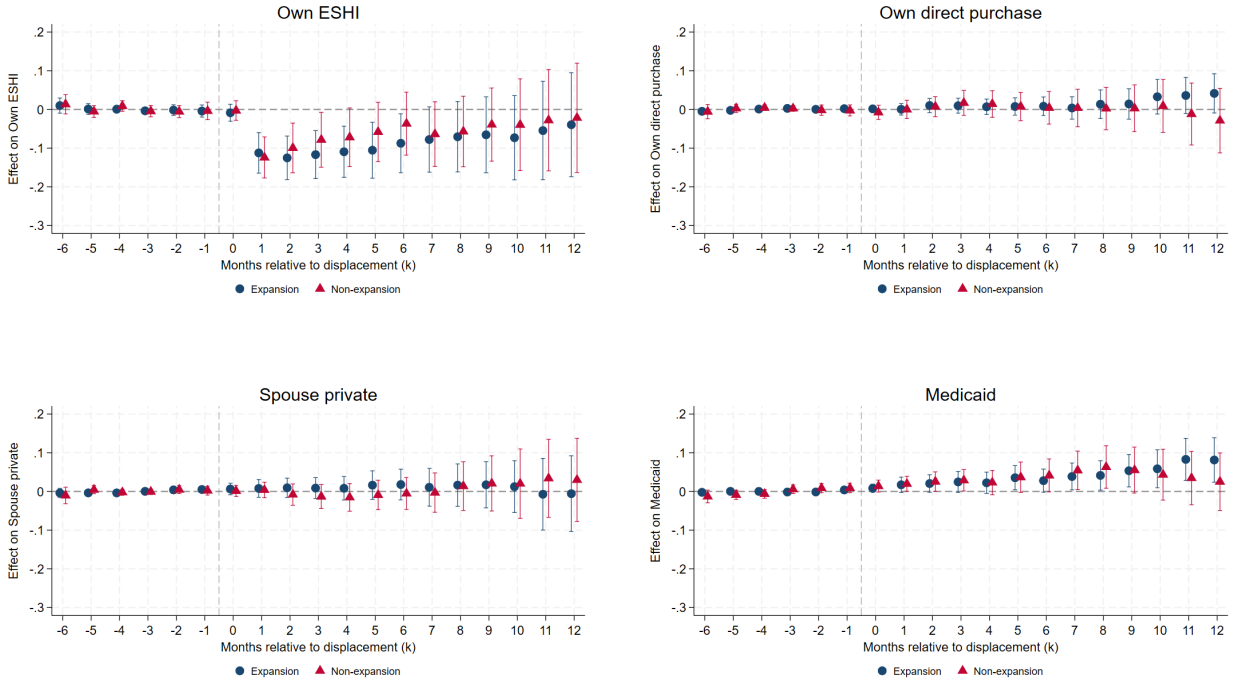
Sample	k	Own ESHI	Own direct purchase	Spouse private	Medicaid
2014 expansion	1	-0.124***	0.001	0.020	0.027**
		(0.036)	(0.010)	(0.015)	(0.013)
	3	-0.148***	0.012	0.019	0.047**
		(0.041)	(0.015)	(0.018)	(0.021)
	9	-0.110*	0.021	0.039	0.090***
		(0.064)	(0.026)	(0.039)	(0.026)
	12	-0.114	0.052	0.053	0.101***
		(0.078)	(0.035)	(0.045)	(0.036)
2014 non-expansion	1	-0.145***	-0.010	-0.002	0.016
		(0.033)	(0.015)	(0.012)	(0.011)
	3	-0.138***	0.008	-0.021	0.027
		(0.038)	(0.020)	(0.020)	(0.017)
	9	-0.059	0.002	0.013	0.021
		(0.055)	(0.037)	(0.046)	(0.030)
	12	-0.027	-0.044	0.028	-0.006
		(0.088)	(0.051)	(0.070)	(0.041)
2018 expansion	1	-0.092**	-0.000	-0.012	0.002
		(0.038)	(0.010)	(0.018)	(0.015)
	3	-0.064	0.005	-0.008	-0.014
		(0.048)	(0.010)	(0.020)	(0.013)
	9	0.034	-0.002	-0.032	-0.029
		(0.072)	(0.026)	(0.046)	(0.038)
	12	0.153	0.015	-0.156	0.030
		(0.129)	(0.023)	(0.131)	(0.051)
2018 non-expansion	1	-0.080*	0.021	0.017	0.027
		(0.046)	(0.020)	(0.020)	(0.019)
	3	0.054	0.038	0.004	0.033
		(0.072)	(0.027)	(0.026)	(0.026)
	9	0.018	0.005	0.044	0.155**
		(0.097)	(0.053)	(0.048)	(0.073)
	12	-0.006	0.015	0.035	0.114
		(0.118)	(0.072)	(0.066)	(0.082)

Panel B: Treated workers contributing to Medicaid estimates

k	2014 expansion	2014 non-expansion	2018 expansion	2018 non-expansion
9	128	123	68	36
12	99	90	50	26

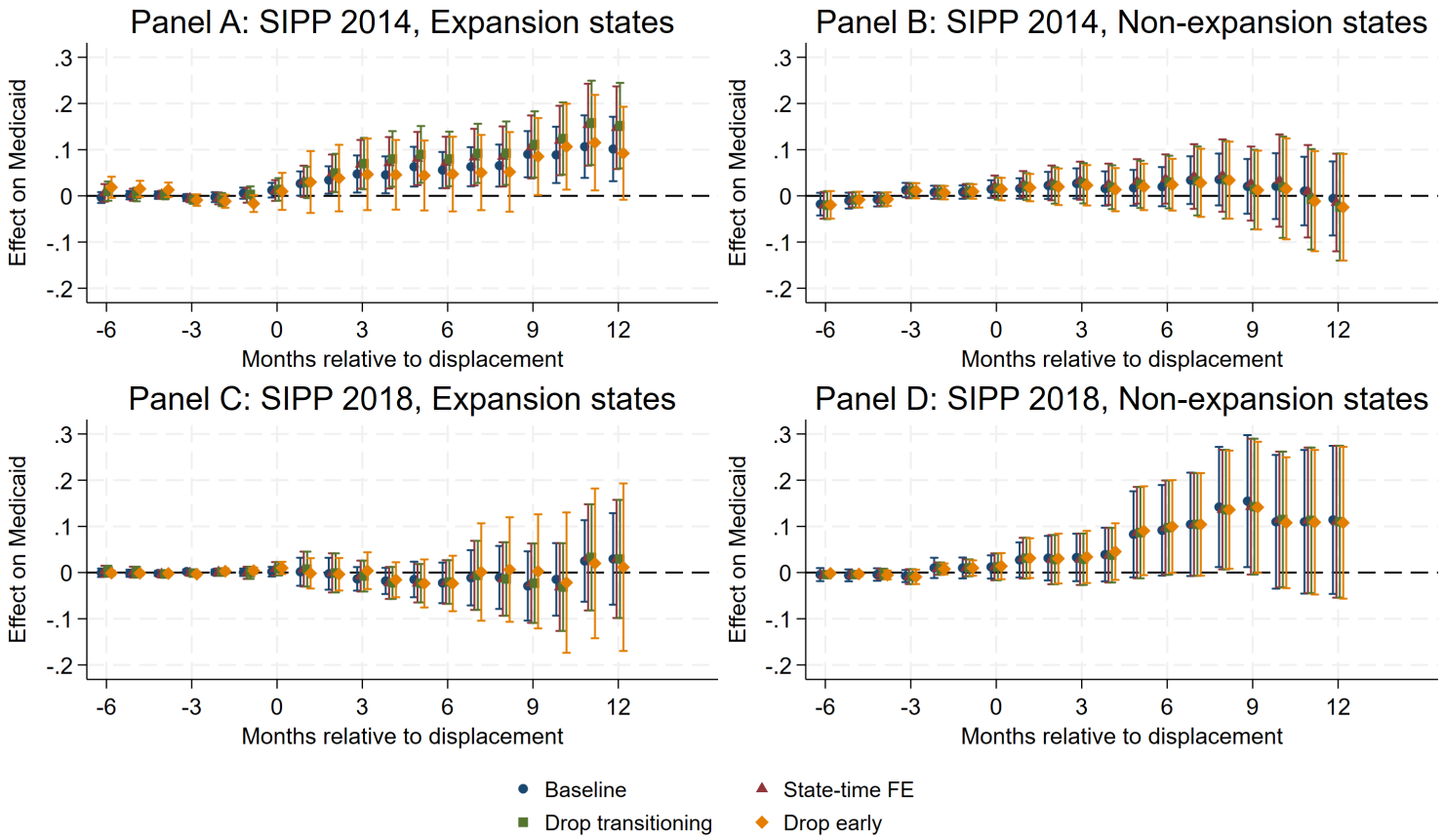
Note: Expansion status is measured at $g-1$. Panel A reports covariate-adjusted coefficients with individual-cluster standard errors in parentheses. Coefficients are expressed as shares. Panel B reports treated-worker support for the later Medicaid estimates discussed in the text. The four insurance-source indicators are not mutually exclusive. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Full event-time paths are shown in Figure 5.

Figure A8: Duration-specific effects of job displacement on health insurance coverage by Medicaid expansion status: pooled post-ACA sample



Note: This figure reports covariate-adjusted event-study estimates by Medicaid expansion status measured at $g-1$ in the pooled post-ACA sample. Estimates use individual survey weights normalized to mean one within each panel. Pointwise 95% confidence intervals are based on standard errors clustered at the individual level. The four insurance-source indicators are not mutually exclusive. Sample and estimation details are described in Sections 2 and 3.

Figure A9: Duration-specific effects of job displacement on Medicaid under alternative specifications



Note: This figure reports covariate-adjusted event-study estimates of the effect of job displacement on Medicaid coverage under the baseline specification, with state-by-month fixed effects, excluding transitioning states, and excluding early-expansion states. Expansion status is measured at $g-1$. Estimates use individual survey weights normalized to mean one within each panel. Pointwise 95% confidence intervals are based on standard errors clustered at the individual level. State-cluster and wild-cluster-bootstrap inference for selected horizons is reported in Table A14.

Table A14: Individual-cluster, state-cluster, and wild-cluster-bootstrap inference for Medicaid effects at selected horizons

Panel A: SIPP 2014, expansion states

k	Statistic	(1) Baseline $G = 48$	(2) State-by-calendar-month FE $G = 36$	(3) Excluding transitioning states $G = 32$	(4) Excluding early-expansion and broader-coverage states $G = 26$
3	Estimate	4.74	6.83	6.99	4.66
	Individual 95% CI	[0.72, 8.76]	[1.55, 12.10]	[1.43, 12.55]	[-3.09, 12.40]
	State 95% CI	[0.12, 9.36]	[2.09, 11.57]	[1.81, 12.16]	[-1.22, 10.53]
	WCU 95% CI		[-0.70, 13.65]		
6	Estimate	5.58	7.23	7.99	4.72
	Individual 95% CI	[1.66, 9.50]	[1.66, 12.80]	[2.09, 13.90]	[-3.37, 12.80]
	State 95% CI	[0.41, 10.75]	[1.90, 12.56]	[2.58, 13.41]	[-0.90, 10.34]
	WCU 95% CI		[-1.63, 14.55]		
9	Estimate	9.02	10.58	11.14	8.51
	Individual 95% CI	[4.02, 14.02]	[3.76, 17.40]	[3.97, 18.31]	[0.18, 16.85]
	State 95% CI	[3.00, 15.05]	[4.07, 17.10]	[4.41, 17.87]	[2.64, 14.39]
	WCU 95% CI		[0.14, 19.60]		
12	Estimate	10.15	14.75	15.21	9.22
	Individual 95% CI	[3.17, 17.12]	[5.81, 23.70]	[5.96, 24.47]	[-0.84, 19.28]
	State 95% CI	[0.30, 19.99]	[4.13, 25.38]	[4.35, 26.08]	[1.93, 16.51]
	WCU 95% CI		[-4.89, 31.92]		

Table A14: Individual-cluster, state-cluster, and wild-cluster-bootstrap inference for Medicaid effects at selected horizons (continued)

Panel B: SIPP 2014, non-expansion states

k	Statistic	(1) Baseline $G = 48$	(2) State-by-calendar-month FE $G = 36$	(3) Excluding transitioning states $G = 32$	(4) Excluding early-expansion and broader-coverage states $G = 26$
3	Estimate	2.68	3.28	2.72	2.29
	Individual 95% CI	[-0.58, 5.95]	[-0.84, 7.40]	[-1.63, 7.08]	[-2.08, 6.66]
	State 95% CI	[0.29, 5.08]	[0.39, 6.17]	[-0.12, 5.57]	[-0.73, 5.31]
	WCU 95% CI		[0.89, 6.69]		
6	Estimate	1.99	3.65	2.98	2.41
	Individual 95% CI	[-2.25, 6.22]	[-1.68, 8.98]	[-2.73, 8.69]	[-3.19, 8.02]
	State 95% CI	[-1.05, 5.02]	[0.53, 6.77]	[-0.15, 6.11]	[-0.76, 5.59]
	WCU 95% CI		[0.95, 7.34]		
9	Estimate	2.05	2.66	1.37	1.25
	Individual 95% CI	[-3.87, 7.98]	[-5.35, 10.68]	[-7.24, 9.98]	[-7.28, 9.78]
	State 95% CI	[-2.05, 6.15]	[-2.20, 7.53]	[-4.04, 6.78]	[-4.14, 6.63]
	WCU 95% CI		[-2.67, 10.07]		
12	Estimate	-0.56	-1.44	-2.41	-2.47
	Individual 95% CI	[-8.58, 7.45]	[-12.01, 9.13]	[-14.02, 9.20]	[-14.02, 9.09]
	State 95% CI	[-4.87, 3.74]	[-5.97, 3.09]	[-7.46, 2.63]	[-7.23, 2.29]
	WCU 95% CI		[-6.78, 5.76]		

Table A14: Individual-cluster, state-cluster, and wild-cluster-bootstrap inference for Medicaid effects at selected horizons (continued)

Panel C: SIPP 2018, expansion states

k	Statistic	(1) Baseline $G = 44$	(2) State-by-calendar-month FE $G = 35$	(3) Excluding transitioning states $G = 35$ Same as (2)	(4) Excluding early-expansion and broader-coverage states $G = 29$
3	Estimate	-1.36	-0.76	Same as (2)	0.41
	Individual 95% CI	[-3.91, 1.19]	[-4.10, 2.59]		[-3.57, 4.39]
	State 95% CI	[-3.58, 0.86]	[-3.52, 2.01]		[-3.65, 4.47]
	WCU 95% CI		[-3.63, 2.42]		
6	Estimate	-2.23	-2.04	Same as (2)	-2.36
	Individual 95% CI	[-6.62, 2.16]	[-6.76, 2.69]		[-8.37, 3.64]
	State 95% CI	[-6.34, 1.87]	[-6.42, 2.35]		[-8.81, 4.08]
	WCU 95% CI		[-6.66, 2.62]		
9	Estimate	-2.90	-2.30	Same as (2)	0.27
	Individual 95% CI	[-10.40, 4.60]	[-10.90, 6.31]		[-12.08, 12.62]
	State 95% CI	[-10.47, 4.67]	[-11.27, 6.67]		[-12.50, 13.04]
	WCU 95% CI		[-11.94, 7.35]		
12	Estimate	2.95	2.95	Same as (2)	1.16
	Individual 95% CI	[-6.96, 12.86]	[-9.84, 15.74]		[-16.98, 19.30]
	State 95% CI	[-5.68, 11.59]	[-8.41, 14.31]		[-18.79, 21.11]
	WCU 95% CI		[-9.98, 14.38]		

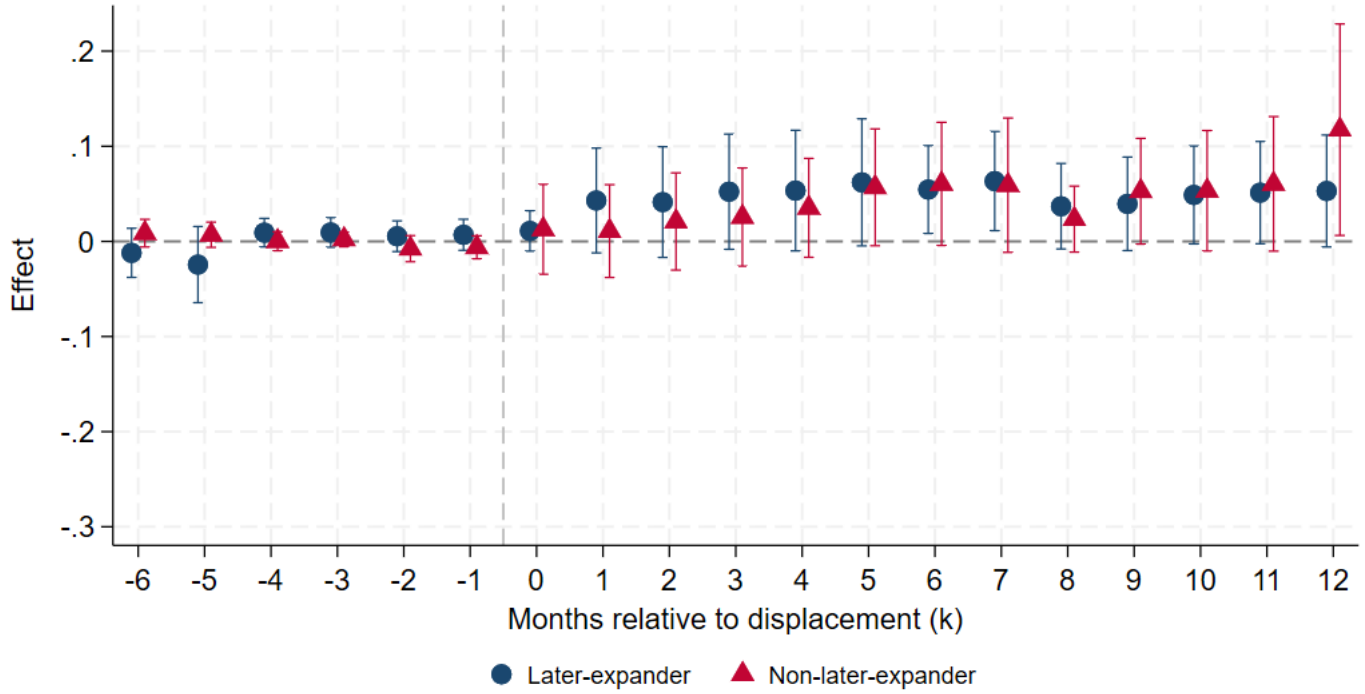
Table A14: Individual-cluster, state-cluster, and wild-cluster-bootstrap inference for Medicaid effects at selected horizons (continued)

Panel D: SIPP 2018, non-expansion states

k	Statistic	(1) Baseline $G = 44$	(2) State-by-calendar-month FE $G = 35$	(3) Excluding transitioning states $G = 35$ Same as (2)	(4) Excluding early-expansion and broader-coverage states $G = 29$
3	Estimate	3.27	2.88	Same as (2)	3.39
	Individual 95% CI	[-1.86, 8.41]	[-2.69, 8.46]		[-2.26, 9.03]
	State 95% CI	[-1.72, 8.27]	[-2.72, 8.49]		[-1.88, 8.65]
	WCU 95% CI		[-3.86, 10.47]		
6	Estimate	9.15	9.73	Same as (2)	9.97
	Individual 95% CI	[-0.65, 18.95]	[-0.48, 19.94]		[-0.07, 20.01]
	State 95% CI	[2.06, 16.24]	[2.48, 16.99]		[3.12, 16.82]
	WCU 95% CI		[1.57, 19.64]		
9	Estimate	15.47	14.28	Same as (2)	14.14
	Individual 95% CI	[1.19, 29.75]	[-0.43, 28.99]		[-0.05, 28.33]
	State 95% CI	[7.47, 23.47]	[4.89, 23.67]		[5.76, 22.51]
	WCU 95% CI		[0.80, 30.22]		
12	Estimate	11.41	11.01	Same as (2)	10.78
	Individual 95% CI	[-4.62, 27.44]	[-5.44, 27.45]		[-5.65, 27.21]
	State 95% CI	[-1.49, 24.31]	[-2.63, 24.64]		[-2.30, 23.87]
	WCU 95% CI		[-11.37, 35.74]		

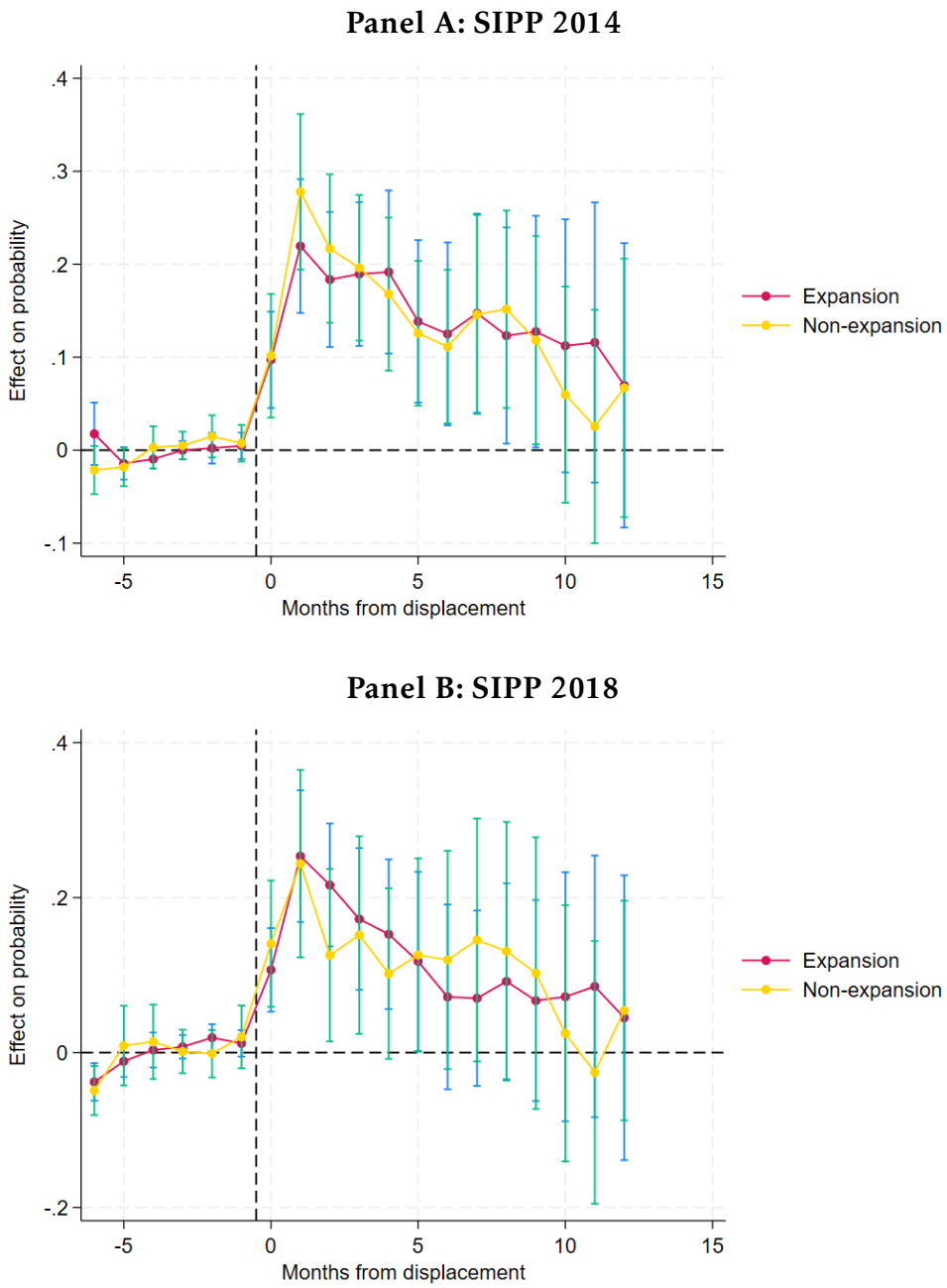
Note: This table reports covariate-adjusted event-study estimates of the effect of job displacement on Medicaid coverage at selected post-displacement horizons. For each event time, the Estimate row reports the point estimate in percentage points. Individual 95% CI denotes the pointwise 95% confidence interval based on standard errors clustered at the individual level and uses the normal approximation. State 95% CI denotes the pointwise 95% confidence interval based on standard errors clustered at the state level and uses the t_{G-1} critical value, where G is the number of states shown in the column heading. WCU 95% CI denotes the pointwise percentile-t 95% confidence interval from a state-level wild-cluster unrestricted bootstrap using 999 Rademacher draws. WCU intervals are reported for column (2). The individual- and state-cluster calculations use the same sample and point estimate within each specification. Expansion status is measured at $g - 1$. In SIPP 2018, no state expanded Medicaid during the estimation window, so column (3) is identical to column (2).

Figure A10: Duration-specific effects of job displacement on Medicaid by later expansion status: pre-ACA sample



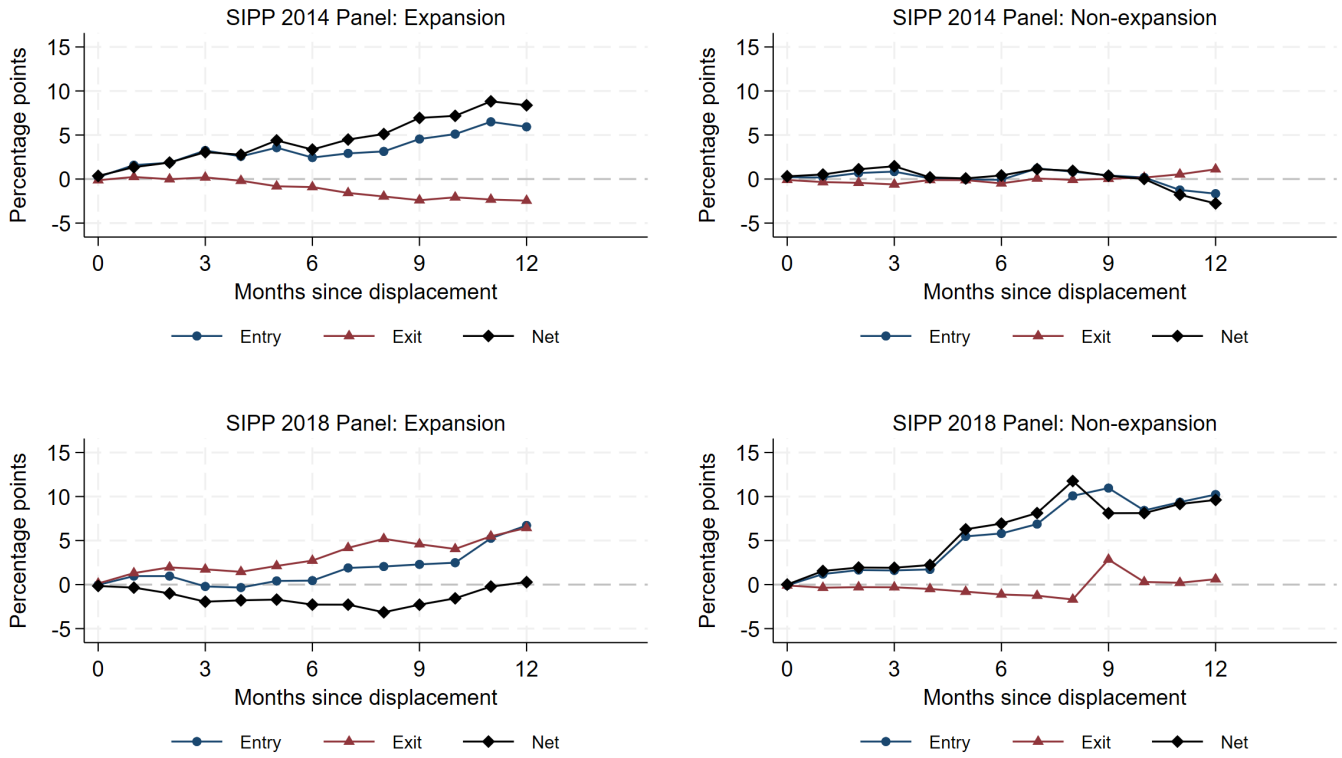
Note: This figure reports covariate-adjusted event-study estimates of the effect of job displacement on Medicaid coverage in the SIPP 2008 panel, separately by later Medicaid expansion status. The later-expander group consists of workers whose state of residence at $g-1$ had adopted Medicaid expansion by December 2019. All other workers are assigned to the non-later-expander group. For workers whose $g-1$ precedes the estimation window, group assignment uses their state of residence in January 2011. The first-stage model includes individual and calendar-month fixed effects and selected baseline covariates interacted with calendar-month indicators. Estimates use SIPP 2008 longitudinal survey weights. Pointwise 95% confidence intervals are based on standard errors clustered at the individual level.

Figure A11: Duration-specific effects of job displacement on family income below 138% of the Census poverty threshold by Medicaid expansion status



Note: Panels A and B report covariate-adjusted event-study estimates of the effect of job displacement on an indicator for monthly family income at or below 138% of the Census poverty threshold in SIPP 2014 and 2018, respectively. Expansion status is measured at $g-1$. Each model is estimated using workers from expansion and non-expansion states and reports separate event-time coefficients for the two groups. Estimates use individual survey weights normalized to mean one within each panel. Pointwise 95% confidence intervals are based on standard errors clustered at the individual level. The measure uses monthly family income reported in SIPP. It is a proxy for satisfying an income requirement because it does not reconstruct the Medicaid eligibility unit, MAGI, or other eligibility rules.

Figure A12: Medicaid entry and exit



Note: This figure decomposes covariate-adjusted event-study point estimates for Medicaid coverage into Entry among workers not enrolled at $g-1$ and Exit among workers enrolled at $g-1$, separately by survey panel and Medicaid expansion status at $g-1$. All components are anchored at $k=-1$. Exit is shown with its estimated sign, and Net equals Entry minus Exit within this decomposition. Net need not coincide with the Medicaid estimate from the main event study because the two analyses use different estimation procedures. Estimates use individual survey weights normalized within panel and are reported in percentage points. Confidence intervals are not computed.

Table A15: Medicaid entry and exit under alternative adjustment sets at selected horizons

		Reduced			Lean		
Panel and group	k	Entry	Exit	Net	Entry	Exit	Net
2014 expansion	1	1.58	0.23	1.35	1.58	0.09	1.49
	3	3.24	0.19	3.05	3.24	-0.08	3.32
	6	2.44	-0.91	3.35	2.44	-1.97	4.41
	9	4.55	-2.40	6.95	4.55	-3.48	8.03
	12	5.93	-2.45	8.38	5.93	-3.88	9.80
2014 non-expansion	1	0.18	-0.34	0.52	0.18	-0.48	0.66
	3	0.85	-0.61	1.46	0.85	-0.71	1.56
	6	-0.08	-0.49	0.41	-0.08	-0.19	0.11
	9	0.41	0.03	0.38	0.41	0.46	-0.05
	12	-1.66	1.11	-2.77	-1.66	1.49	-3.15
2018 expansion	1	0.96	1.31	-0.34	0.96	0.89	0.08
	3	-0.21	1.73	-1.94	-0.21	0.68	-0.88
	6	0.45	2.73	-2.27	0.45	1.66	-1.21
	9	2.30	4.58	-2.28	2.30	4.23	-1.93
	12	6.72	6.44	0.28	6.72	3.30	3.42
2018 non-expansion	1	1.20	-0.35	1.54	1.20	-0.12	1.32
	3	1.61	-0.30	1.91	1.61	-0.13	1.74
	6	5.81	-1.12	6.93	5.81	-1.01	6.82
	9	10.96	2.85	8.11	10.96	-2.29	13.25
	12	10.23	0.61	9.62	10.23	-0.04	10.27

Note: Entry and exit are defined relative to Medicaid enrollment at $k = -1$ and are reported in percentage points. Net equals entry minus exit. For baseline enrollees, reduced adjustment omits baseline spouse full-time employment in 2014 and baseline own employment and earnings in 2018, together with their calendar-month interactions. Lean adjustment removes all baseline covariate-by-month interactions, retaining individual and calendar-month fixed effects and linear age in 2018. For workers not enrolled in Medicaid at baseline, both specifications include the same covariates as the main Medicaid analysis. Confidence intervals are not computed. Full event-time paths for the reduced specification are shown in Figure A12.

Table A16: Duration-specific effects and treated-worker support: pre- and post-ACA samples

Panel A: Event-study estimates

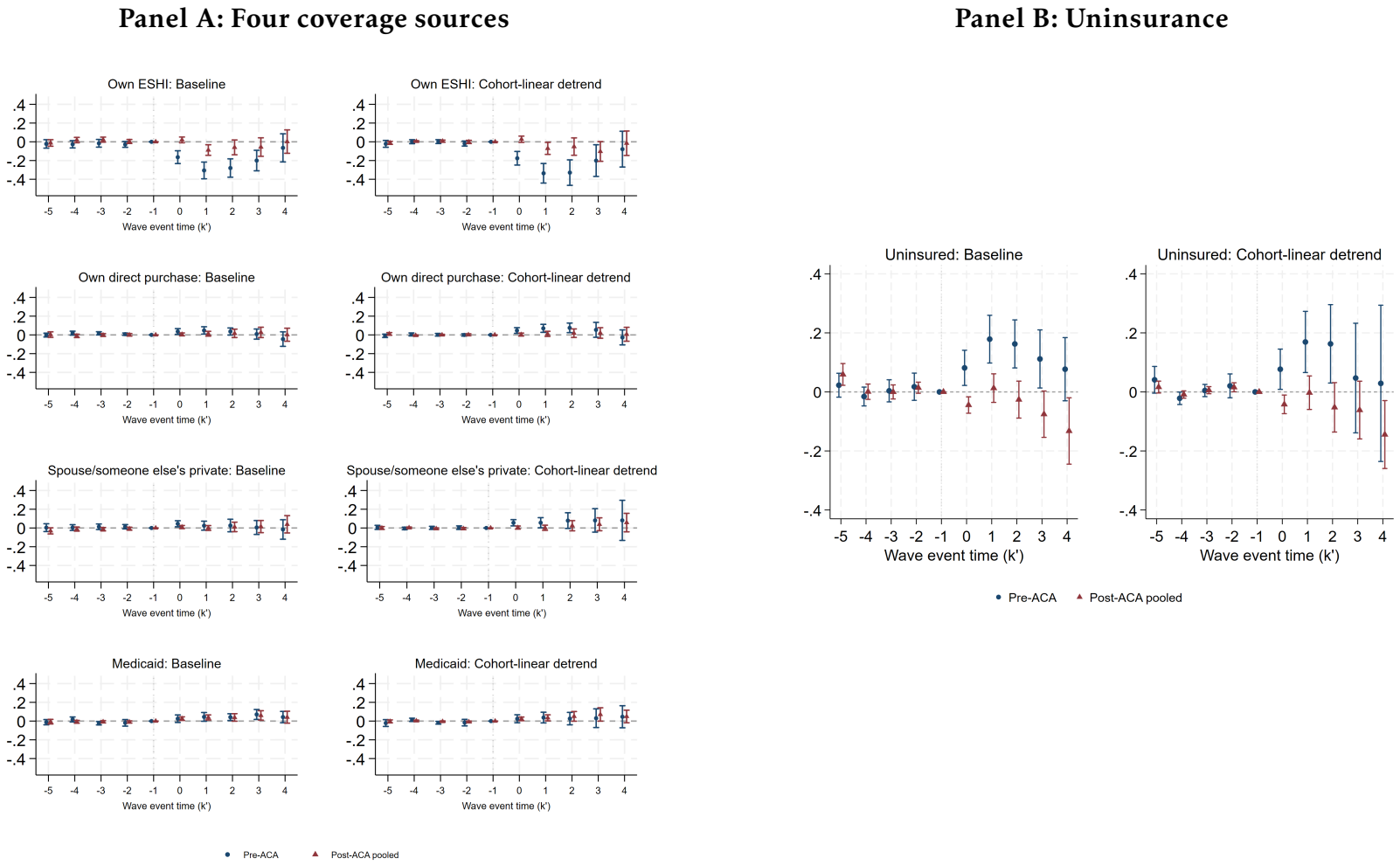
Outcome	k'	Rebased pre-ACA	Rebased post-ACA	Pre–post gap
Own ESHI	0	-0.164 (0.035)	0.020 (0.016)	-0.184 (0.038)
	1	-0.306 (0.046)	-0.088 (0.029)	-0.218 (0.054)
	2	-0.280 (0.050)	-0.059 (0.040)	-0.221 (0.064)
	3	-0.200 (0.056)	-0.056 (0.050)	-0.144 (0.075)
	4	-0.065 (0.077)	0.002 (0.064)	-0.067 (0.100)
Own direct purchase	0	0.036 (0.016)	0.005 (0.008)	0.031 (0.017)
	1	0.047 (0.020)	0.012 (0.013)	0.035 (0.024)
	2	0.036 (0.019)	0.016 (0.023)	0.020 (0.030)
	3	0.008 (0.028)	0.027 (0.028)	-0.018 (0.039)
	4	-0.045 (0.039)	0.002 (0.036)	-0.047 (0.053)
Spouse/someone else's private	0	0.045 (0.016)	0.010 (0.009)	0.035 (0.018)
	1	0.025 (0.025)	-0.000 (0.013)	0.025 (0.028)
	2	0.026 (0.034)	0.011 (0.026)	0.015 (0.043)
	3	0.005 (0.038)	0.015 (0.033)	-0.010 (0.050)
	4	-0.016 (0.053)	0.039 (0.047)	-0.055 (0.071)
Medicaid	0	0.025 (0.021)	0.026 (0.010)	-0.001 (0.023)
	1	0.045 (0.024)	0.039 (0.014)	0.006 (0.028)
	2	0.042 (0.018)	0.039 (0.020)	0.003 (0.027)
	3	0.071 (0.027)	0.060 (0.026)	0.010 (0.037)
	4	0.043 (0.031)	0.041 (0.033)	0.001 (0.045)
Uninsured	0	0.081 (0.030)	-0.044 (0.014)	0.126 (0.034)
	1	0.179 (0.041)	0.013 (0.025)	0.166 (0.048)
	2	0.163 (0.042)	-0.026 (0.032)	0.189 (0.052)
	3	0.112 (0.050)	-0.075 (0.040)	0.187 (0.064)
	4	0.077 (0.055)	-0.132 (0.057)	0.209 (0.079)

Panel B: Treated workers contributing at each event time

	Treated workers		
k'	SIPP 2008	SIPP 2014	SIPP 2018
−6	18	–	–
−5	56	58	46
−4	109	129	106
−3	156	209	143
−2	204	266	165
−1	259	346	197
0	259	346	197
1	230	288	151
2	189	217	91
3	138	137	54
4	93	80	32
5	48	–	–

Note: Panel A reports covariate-adjusted wave-level event-study estimates for the pre- and post-ACA samples. The pre-ACA series uses SIPP 2008 waves. The post-ACA series uses four-month bins from pooled SIPP 2014 and 2018. Estimates are recentered by subtracting the coefficient at $k' = -1$, and the gap is pre-ACA minus post-ACA. Coefficients and individual-cluster standard errors in parentheses are expressed as shares. Gap standard errors equal the square root of the sum of the two squared standard errors because the panels are independent. The pre-ACA someone-else private outcome is paired with post-ACA spouse private coverage. Panel B reports the number of treated workers contributing to each event-time coefficient. Support is identical across the six insurance outcomes and is shown once for each SIPP panel. Workers displaced in their first observed wave are excluded.

Figure A13: Duration-specific effects under baseline and detrended specifications



Note: The figure compares baseline covariate-adjusted and cohort-linearly detrended wave-level estimates for the pre- and post-ACA samples. The pre-ACA series uses SIPP 2008 waves. The post-ACA series uses four-month bins from pooled SIPP 2014 and 2018. Pointwise 95% confidence intervals use standard errors clustered at the individual level. All series are recentered at $k' = -1$.

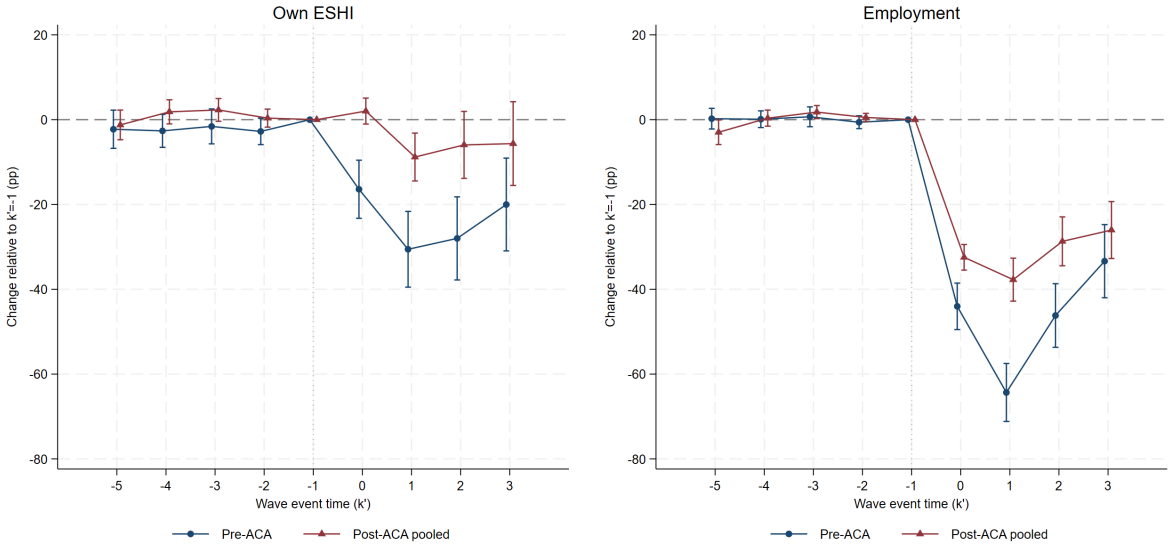
Table A17: Pre- and post-ACA effects before and after detrending

Outcome	Specification	Pre-ACA	Post-ACA	Gap (pre–post)
Own ESHI loss	Baseline	30.6	8.8	21.8
	Detrended	33.6	7.0	26.6
Uninsurance increase	Baseline	17.9	1.3	16.6
	Detrended	16.9	-0.3	17.2

Note: Point estimates are for the wave following displacement ($k' = 1$), relative to $k' = -1$, in percentage points. Own ESHI is reported as a decline and uninsurance as an increase. Detrended estimates include cohort-specific linear trends fitted using only untreated observations. Gaps are calculated before rounding. Figure A13 shows the full paths.

Figure A14: Duration-specific effects on own ESHI and employment: pre- and post-ACA samples

Panel A: Event-study estimates

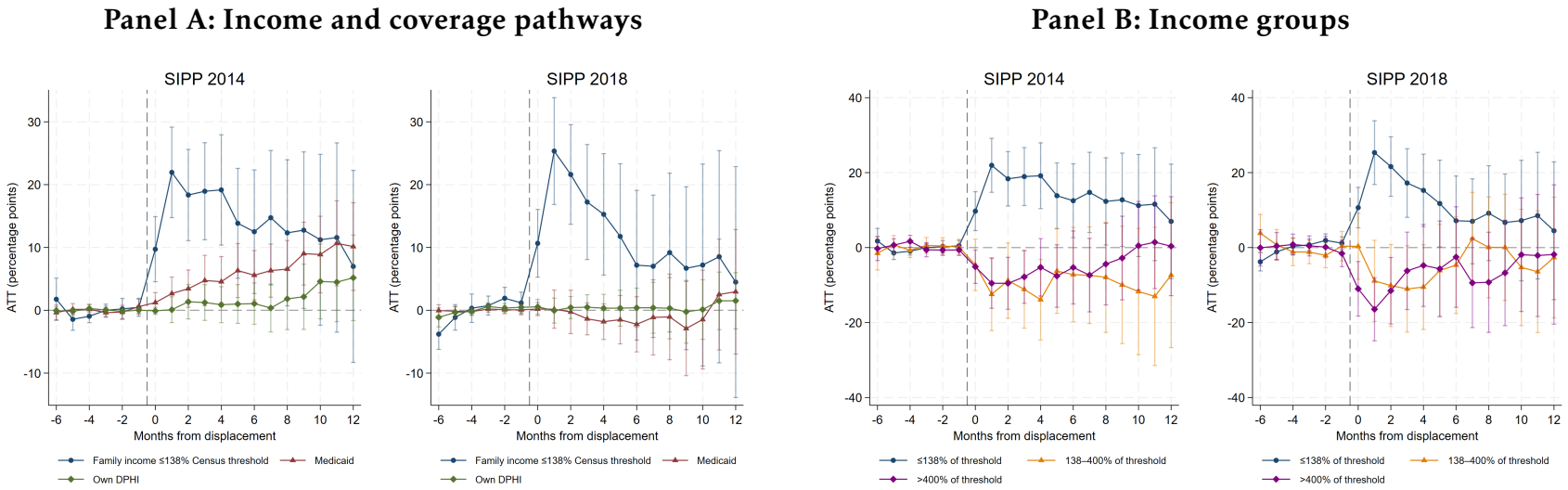


Panel B: Own ESHI declines relative to employment declines

	Pre-ACA			Post-ACA pooled		
Event time	Own ESHI (pp)	Employment (pp)	Ratio	Own ESHI (pp)	Employment (pp)	Ratio
$k' = 1$	30.57	64.34	0.475	8.81	37.73	0.234
$k' = 2$	28.00	46.19	0.606	5.95	28.69	0.207
$k' = 3$	20.01	33.37	0.600	5.64	26.03	0.217

Note: Panel A reports baseline covariate-adjusted wave-level event-study estimates without detrending. The pre-ACA series uses SIPP 2008 waves, including partial waves at the analysis-window edges. The post-ACA series uses four-month bins from pooled SIPP 2014 and 2018. Estimates use the longitudinal survey weights specified for each panel, with post-ACA weights normalized to mean one within panel. Pointwise 95% confidence intervals use standard errors clustered at the individual level. Estimates are recentered at $k' = -1$. Own ESHI measures coverage at any time within a wave, and employment is the share of observed months employed. Panel B reports declines in percentage points and their ratios. The ratios are descriptive ratios of estimated effects. Confidence intervals for the ratios and tests of differences between ratios are not reported.

Figure A15: Duration-specific effects on family income position and coverage pathways: Medicaid expansion states



Note: Panel A reports covariate-adjusted event-study estimates for monthly family income at or below 138% of the Census poverty threshold, Medicaid coverage, and own direct-purchase coverage. Panel B reports estimates for three mutually exclusive and exhaustive income groups: at or below 138%, above 138% and at or below 400%, and above 400% of the Census poverty threshold. Each display reports SIPP 2014 and 2018 in its left and right panels. The sample is limited to workers in Medicaid expansion states, with expansion status measured at $g-1$. The income indicators describe current income position and do not fully measure Medicaid or Marketplace subsidy eligibility. Event-time coefficients for $k=-6,\ldots,12$ are reported in percentage points and are not recentered at $k=-1$. Estimates use individual survey weights normalized to mean one within each panel. Pointwise 95% confidence intervals use standard errors clustered at the individual level. Because the income groups partition the sample, their coefficients sum to zero at each event time.